

User's Guide



Version # 4.4.6

Princeton Instruments

How to contact us:

Telephone: (978) 263-3584

Fax: (978) 263-5086

E-Mail: mail@piacton.com

Internet: <http://www.piacton.com>

Mailing Address: Princeton Instruments - Acton

15 Discovery Way

Acton, MA 01720

Street Address: Princeton Instruments - Acton

15 Discovery Way

Acton, MA 01720

Copyright © 1999-2007 Acton Research Corporation. All rights reserved.

Copyright © 2008 Princeton Instruments. All rights reserved.

Excel, Microsoft, Windows, Windows NT are registered trademarks of Microsoft Corporation in the United States and/or other countries.

GRAMS, GRAMS/32, and GRAMS/3D are registered trademarks of Thermo Lab Systems Inc.

No part of this document may be reproduced by any means without the written permission of Princeton Instruments.

Information in this manual is subject to change without notice.

Table of Contents

Chapter 1: Introduction	5
General Overview of SpectraSense.....	5
Hardware Config	5
Survey Mode	5
Acquisition	5
Live Data	6
Post Processing.....	6
Expanding Spectra and Images	6
File Formats	6
Software Installation	6
Hardware Installation.....	7
Chapter 2: Hardware Config Screen.....	9
Optics tab	11
CCD tab	12
Detection tab	14
SpectraHub or NCL.....	14
SR-510/SR-810	16
Monochromator tab.....	21
Filter Wheel tab.....	22
Shutters tab	23
NCL Triggers tab	24
CCD Triggers tab	25
Printer Options tab	26
Menu Options	27
Chapter 3: CCD Survey Screen.....	29
CCD Survey Mode.....	29
Main Display Window and Functions.....	29
Acquisition Mode and Adjustments buttons.....	29
Cursor Coordinates panel	30
Adjustments tabs	30
Setup tab	30
Display tab.....	31
Area Maps tab	32
Mono1 tab	33
Mono2 tab	33
CCD Gain tab	34
Lower Display Window and Functions.....	34
Imaging Mode	34
Spectral Mode.....	35
Real Time Processing panel.....	35
Filter panel.....	36
Trigger Status panel.....	36
Cosmic Correction and Dark Subtraction panels	36
3d/Log panel.....	36
Cursor/Area Selection panel.....	37
External Sync panel.....	37
Calibration	37

Menu Options	39
Chapter 4: Scanning Survey Screen	41
Scanning Survey Mode	41
Scan Range Selection panel	41
Adjustments button.....	42
Setup tab	42
Det HV tab.....	42
Mono tab	43
Real Time Processing panel.....	43
Go/Stop button	44
Optimize/Stop button	44
Clear button	44
Cursor Information panel	44
Display Format options	44
Show Half Width check box.....	45
Shutter and Filter panels.....	45
Trigger Status panel.....	45
High Voltage table	45
Menu Options	46
Chapter 5: CCD-Based Acquisition	49
CCD Spectral Acquisition.....	49
Acquisition Parameters	50
Area Map Information.....	50
CCD Spectral Position panel.....	50
Readout panel.....	51
Real Time Processing panel.....	51
Filter Wheel panel	52
Format/Storage panel.....	52
Excitation Profile panel.....	53
File Parameters.....	54
Routine panel.....	54
Settings panel	54
Operator/Comments panel.....	55
Save panel	55
Menu Options	56
Chapter 6: Scanning Spectral Acquisition	57
Single Point Scan Mono	57
Acquisition Parameters	58
Scanning Parameters panel.....	58
Data Acquisition panel	58
High Voltage panel	58
Real Time Processing panel.....	59
Excitation Profile panel.....	60
Filter Wheel panel	60
Format/Storage panel.....	60
File Parameters.....	61
Routine panel.....	61
Settings panel	61
Operator/Comments panel.....	62

Save panel	62
Menu Options	63
Chapter 7: Scanning Time-Based Acquisition	65
Single Point Intensity versus Time Acquisition	65
Acquisition Parameters	66
Time panel	66
Wavelength panel	66
Real Time Processing panel.....	66
High Voltage panel	67
Filter Wheel panel	67
Format/Storage panel.....	67
File Parameters.....	68
Routine panel.....	68
Settings panel	68
Operator/Comments panel.....	69
Save panel	69
Menu Options	70
Chapter 8: Live Data Screen	71
Live Data Display	71
CCD Mode Display Options	71
Cycles/Areas.....	71
Scaling.....	72
Scanning Mode Display Options	72
Cursor panel	72
Scaling.....	72
Stop Button	72
Header/Comment	73
Comments panel.....	73
Comment Form dialog.....	73
Plot Formatting Tools.....	73
Menu Options	74
Chapter 9: Post Processing Screen	75
Mathematical Operations	76
History	78
Menu Options	79
Chapter 10: Importing Data into GRAMS®	81
Glossary	83
Appendix A.....	85
Focusing Your Princeton Instruments CCD Detector mounted on an Acton Spectrometer.....	85
Appendix B.....	87
Improving the signal to noise ratio of your CCD spectra	87
Appendix C.....	89
Trigger order of operation.....	89
Appendix D.....	91
Image Tiff 6.0 Compliance	91

Index	95
--------------------	-----------

Chapter 1: Introduction

General Overview of SpectraSense

SpectraSense is a comprehensive spectral acquisition and data treatment package. It is designed to work exclusively with Princeton Instruments Acton series instrumentation and accessories. It runs under Windows® 2000, Windows NT®, and Windows XP operating systems (with the exception of *InSpectrum* or USB-based instrumentation which can only run under Windows 2000 and Windows XP). SpectraSense software will control and acquire data from both single channel detection systems, which require scanning of a monochromator, and CCD based detection systems that may or may not require repositioning of a spectrograph. Depending upon the hardware purchased, either one or both of the modes of acquisition will appear during normal operation.

SpectraSense is presented as a series of tabbed screens. Depending on the desired operation, you may jump between any tabs. The tabs are presented in a logical order for acquiring and interpreting quality spectra. All tabs will be accessible as long as there is hardware to support their operation. A brief discussion of the purpose and operational capabilities of each tab follow.

Hardware Config

The Hardware Config screen is used to configure the software to recognize and operate with the various automated components of your system. It is in this screen that you tell the software what you have and where you have put it. A diagrammatic representation of your system is created by placing icons that represent components and accessories on a monochromator diagram. At a glance you can verify or change gratings, adjust automated slits, change from one mode of detection to another, verify or change the position of your spectrometer(s), and change the states of your triggers, filters, high voltage, and shutters.

Survey Mode

The Survey Screen is used to quickly set up and optimize the parameters that affect the quality of your spectra. In real-time you can adjust factors such as integration time, slit width, high voltage, areas on a CCD detector and spectrometer positioning. When working with a CCD you can evaluate the signal as either an image or a spectrum. With a single channel detection system you can sit on a peak and watch how the signal level varies as you change operating parameters.

You can do real-time manipulation of incoming data and see the effects in varying parameters associated with one detector relative to another. And, finally you can store the spectra that you have taken here. All of the optimization and tweaking that you have done in this screen is automatically transferred to the Acquisition screen so that you can be sure that the spectra being collected are as good as you can get.

Acquisition

In the Acquisition screen, you define how your experiment is to be performed. It is here that you create routines that can be recalled at any time to exactly reproduce the experimental conditions used in a previous session. It is also here that you can add anecdotal information to be stored with the data. It is from this screen that you acquire your spectral data.

Live Data

The Live Data screen shows the data as it is being acquired. In certain high speed acquisitions using CCD Detectors however, it may not be possible to see the data until it has been collected.

Post Processing

After you have collected the data, you can view it, compare it to other data, and perform mathematical manipulations on it. It is also from this screen that you can create hard copies of the spectra.

Expanding Spectra and Images

All screens, windows, and dialog boxes that contain spectra can be expanded. To expand around a section of a spectrum draw a box around the section by clicking the left mouse button and dragging the cursor from the top left corner of the section to the bottom right. To return to full scale expansion, click and drag the cursor anywhere on the display from bottom right to top left. You can move an expanded display left, right, up, or down by clicking and holding the right mouse button. Images taken with the CCD can be expanded only if they are in the primary display.

File Formats

Spectral data files can be stored in one of three formats: Spectrum.**arc_data**, Spectrum.**txt** or Spectrum.**SPC**. The **arc_data** format contains all of the acquisition, instrumentation, and data manipulations information in a text header as well as the data. This is an ASCII format. The **.txt** format contains only the data as X and Y coordinates in a tab delimited ASCII file. The **SPC** format is a proprietary file format used by Galactic Industries' GRAMS[®] software. Data stored in this format can be directly imported into GRAMS. You select the storage format in the Save dialog box in the Save as choice field. Use the drop arrow to select the format you want to use.

Images can be stored as a Windows bitmap (*.**BMP**). They can also be saved as in a Galactic (*.**SPC**) format which will import the image into GRAMS as a series of spectra, one for each row of the image. Two **Tiff** formats are also available, 8.bit RGB and 16 bit gray scale. Select the format you want to save in by clicking on the drop arrow in the Save as field in the Save dialog box.

Software Installation

In order to install SpectraSense, you must have a computer with at least a Pentium processor or equivalent, 64 Mb of RAM or more, a hard disk drive with 200 Mb of available space, a SVGA or better graphics card, a monitor, a CD ROM reader, a Windows-compatible mouse and a Windows-compatible printer. The operating system must be Windows NT, Windows 2000, or Windows XP with the exception of **InSpectrum** and USB-based instrumentation which can only run under Windows 2000 or Windows XP. Depending on the Windows version, you may also be required to have Administrator's rights.

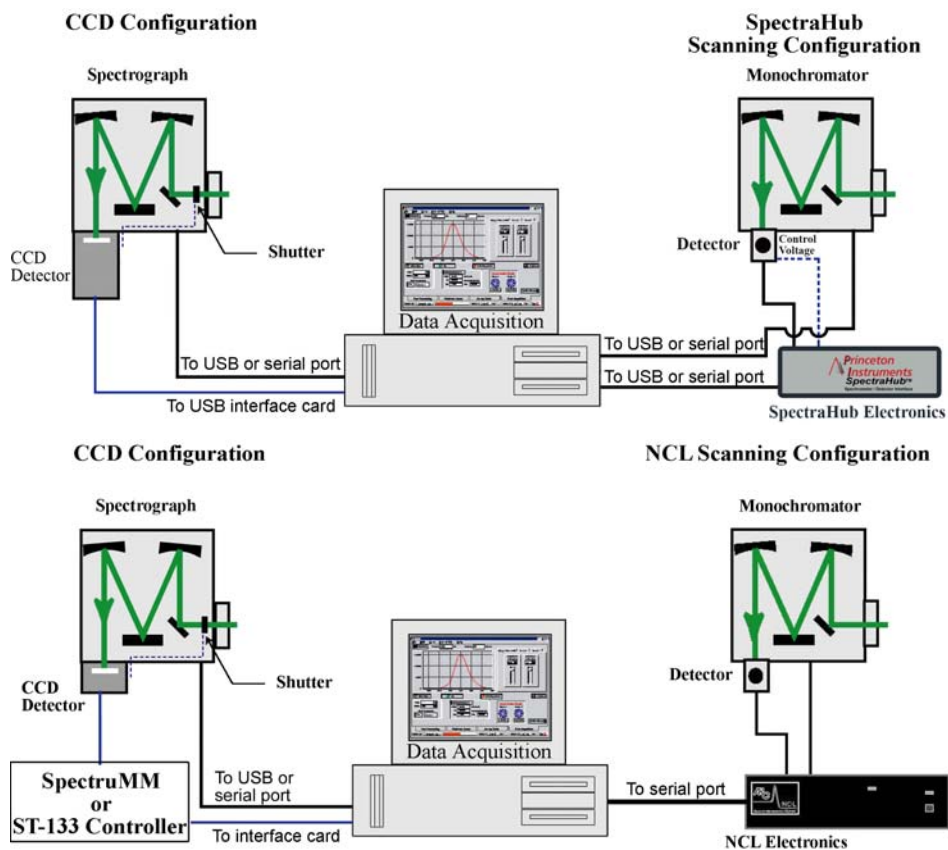
Place the SpectraSense CD-ROM in the drive. Click on the **Start** button, go up to **Run**. Click on the **Browse** button and find the drive and directory that contains Autorun.exe. Click on **Autorun.exe** and then **Open**. Click on **OK**.

An installation panel will appear. Click on the install button that is appropriate for your operating system. Follow the directions on the screen to complete the installation. Your computer must be rebooted before you can use the software.

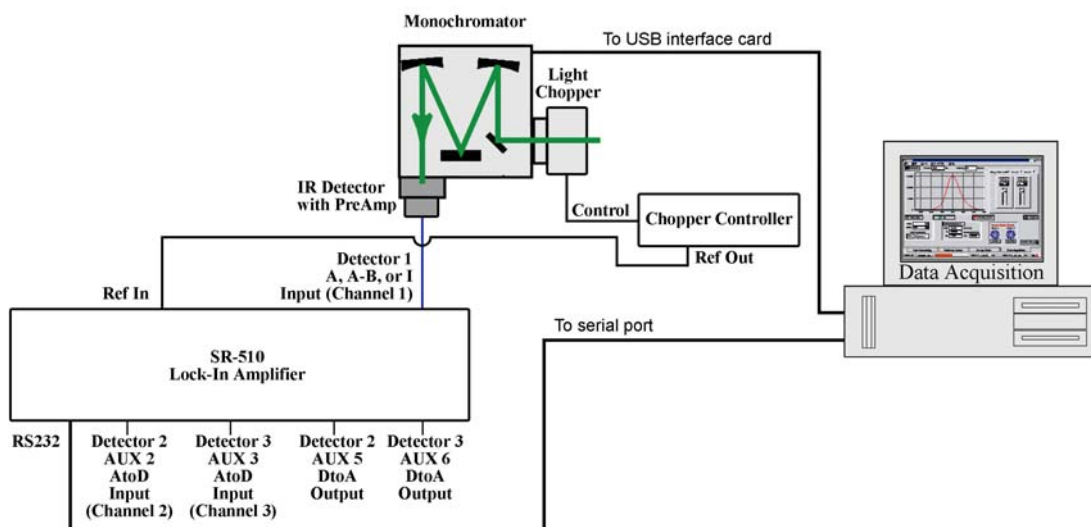
The complete manual in PDF format is also included on the installation CD-ROM. All topics in the table of contents and index are hyperlinked to the appropriate pages in the manual. In order to read the manual, you must have an Adobe Acrobat Reader on your computer. If you do not have it, click on the install Adobe Acrobat Reader button. A complete catalog of Acton series products and technical notes are included on the CD-ROM.

Hardware Installation

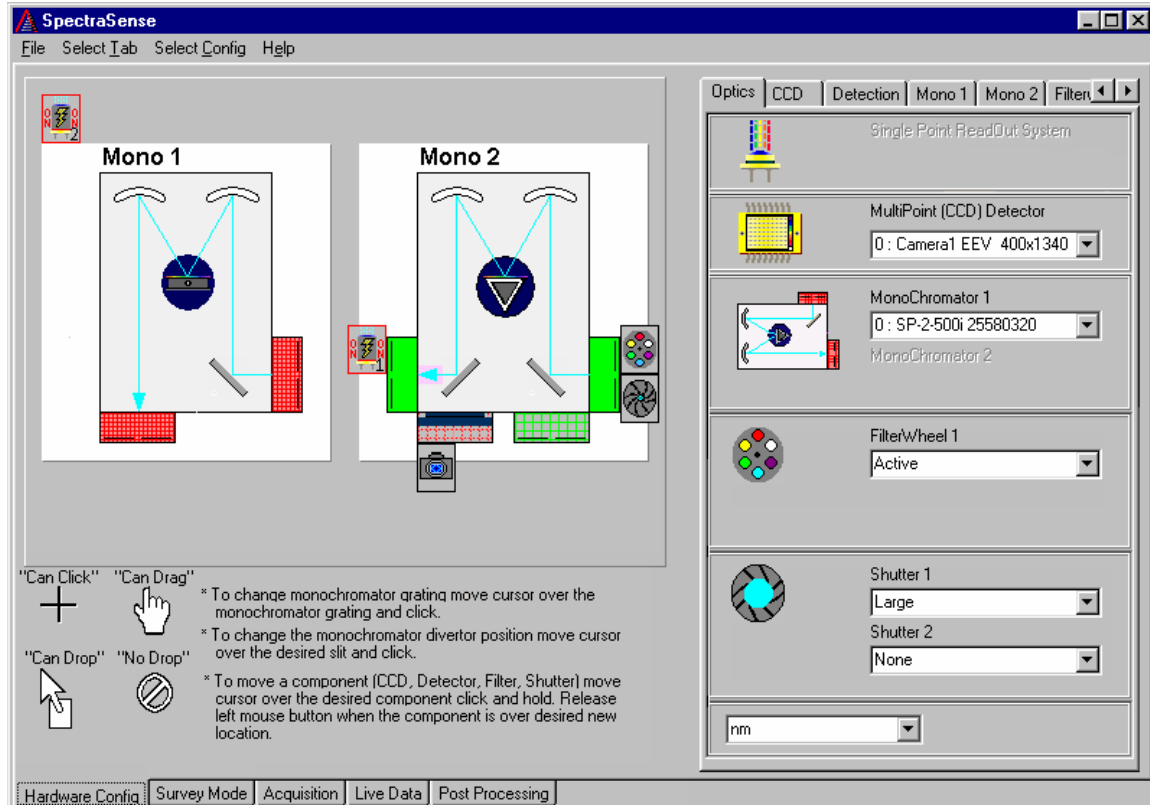
Your hardware must be set up and all of the appropriate cables attached before the software can work. Refer to each component's installation manual for specifics. The diagram below shows typical cabling configurations. *InSpectrum* has only one connection between it and the computer through the USB port. See the *InSpectrum* manual for further details.



SR-510 Configuration



Chapter 2: Hardware Config Screen

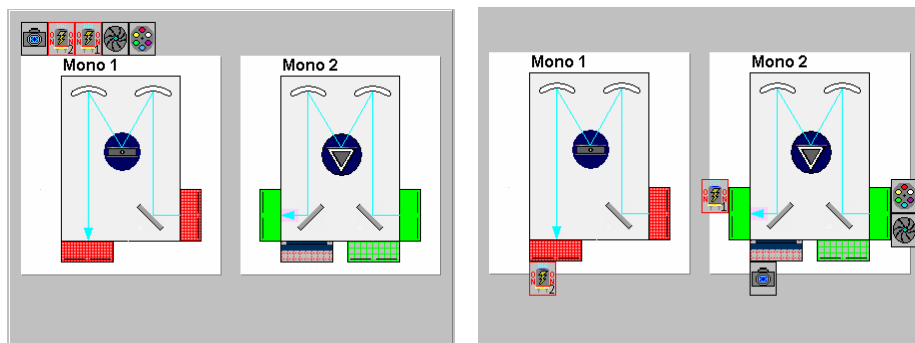


The Hardware Config screen is used to turn on or off software control features related to the specific hardware in your experimental setup. In principle, the diagram represents the physical placement of all of the software-controllable components in your system. Once the system is configured, there should be no need to return to this tab unless you want to reconfigure the physical system or you want to review the current hardware setup.

The system diagram to the left is updated with icons for the specific hardware included in your system as you activate components from the panel to the right. Monochromators, the NCL data acquisition system, SpectraHub data acquisition system, and CCD detector systems are automatically detected when the system is turned on. Shutters and filter wheels must be specified as being active on the OPTICS tab. The SR-510 data acquisition system must be activated via the Select Config menu and SpectraSense must then be restarted.

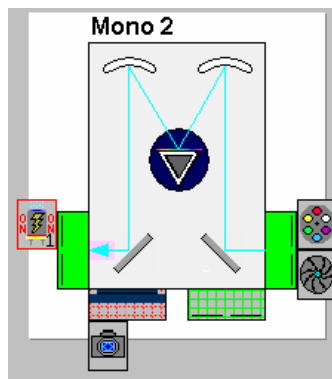
In general, as components are activated through the tabs on the panel, icons are created and placed on the top of the system diagram. You then place the icons in the appropriate position in the experimental set up. (Click on an icon, drag the cursor to the appropriate port and release.) In this way, the software associates the specific components to specific ports on the monochromator(s). Components assigned to ports that are not in the current light path will be disabled and all options associated with them hidden.

Note: When a Stanford Research 510 lock-in amplifier (SR-510) is part of the system, it must be activated by selecting Configure SR-510 Lock-In from the Select Config menu. Then, in the SR-510 dialog, you must indicate the Com Port, check "Use SR-510 Lock-In", click on OK, and restart SpectraSense before the lock-in icons are displayed.



The optical layout with the current optical path is shown on the system diagram. If there are two spectrometers, both will be shown and you make the Mono1 and Mono2 assignments with the Optics Tab MonoChromator 1 and MonoChromator 2 drop downs. The layout to the left shows the light entering from the side entrance slit and going to the CCD camera. The optical path can be changed in either of two ways; by clicking on the turning mirrors or by clicking on the desired entrance and exit ports. The blue optical path will change to reflect the selected entrance and exit ports.

The current grating can be verified or changed by clicking on the grating turret. A drop down menu with all of the available gratings will appear. Solid green slits are motorized and software-controllable. Solid red slits are manual. Checkered slits are out of the optical path.



The following tabs may show up on the panel depending upon the hardware specified in your system:

- OPTICS** Shows monochromators present, CCD if present, and allows activation of filter wheels and shutters.
- CCD** This tab appears only if a CCD detector system is detected.
- DETECTION** This tab appears if an SR-510, a SpectraHub, or an NCL is present and is used to configure the system for an SR-510 (or compatible) lock-in amplifier and for single channel detectors.
- MONO 1** This tab appears if a monochromator or an *InSpectrum* is detected
- MONO 2** This tab appears if a second monochromator is detected (requires a SpectraHub, an NCL controller, or more than one monochromator)
- FILTER WHEEL** This tab appears if a filter wheel has been specified as being active on the OPTICS tab.
- SHUTTERS** This tab appears if a shutter is specified on the OPTICS tab.
- NCL TRIGGERS** This tab appears if a SpectraHub or an NCL is present. It is used to configure the triggering conditions and actions that start an acquisition or terminate it.
- CCD TRIGGERS** This tab is used to configure the CCD triggers.
- PRINTER** This tab is used to configure printing parameters.

Optics tab

The optics tab shows the found monochromator(s), single point readout systems, and CCD detection systems. It is here that you specify if there are filter wheels and shutters in your system. You also specify your working spectral units from this tab:

Single Point Readout System: List of single point detectors detected by the software. If the system has an SR-510, it must be configured and the software must be rebooted before the SR-510 will appear in this panel.

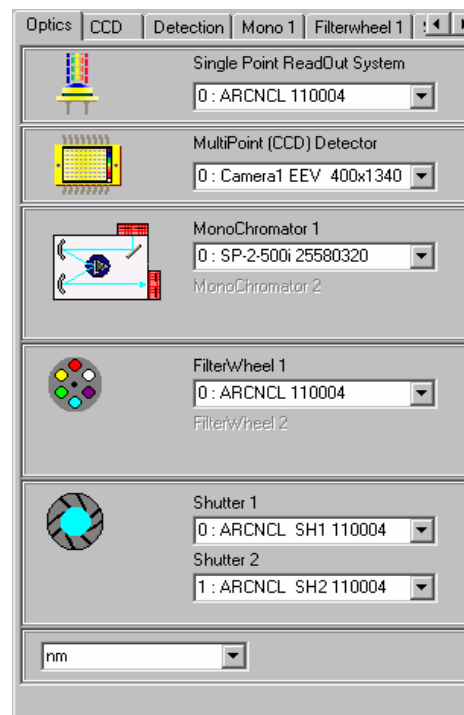
MultiPoint (CCD) Detector: List of PVCAM-supported camera(s) detected by the software.

MonoChromator: List of spectrometers detected by the software. If more than one spectrometer has been detected, there will be two dropdown lists: MonoChromator 1 and MonoChromator 2. Use these dropdown lists to specify which monochromator is Mono1 and which is Mono2.

Filter Wheel: Lists the filter wheels detected. Select a filter wheel or select "Do Not Use". If you select "Do Not Use", the associated Filterwheel tab will disappear.

Shutters: Lists the shutters detected. Select a shutter or select "Do Not Use". If there are two shutters, and you select "Do Not Use" for both, the Shutter tab will disappear.

Working spectral units: Select from nm, microns, EV, Angstroms, absolute wavenumbers or relative wavenumbers with a laser wavelength specified in nm. All displacements of the monochromators will be in the specified units. If motorized slits are present, slit width adjustments can be specified in band passes of the selected working unit.



CCD tab

This tab appears if a CCD detector is found by the software. The settings and parameters on this tab will change depending on the detector. For example, Gain Multification will only appear if the detector uses EMCCD technology.

Note: If there is a Set button associated with a parameter, click on that button after you have made your choice.

CCD Detector: The drop down lets you select from the PVCAM-compatible detectors (up to eight) found by PVCAM. If you have one detector, there will be only one drop down item, if you have eight, there will be eight items. Select the model that you are using for this setup.

Head Temperature: Set the head temperature to the lowest value specified for that model. If the minimum and maximum temperatures can be read from the detector, this information will be reported. Otherwise, refer to the detector documentation for the appropriate value.

Analog to Digital Converter:

- **Ports:** If there are two readout ports, select the one that meets your needs: low noise if your signal has a small signal to noise (SNR) ratio or high capacity if there is a high SNR.
- **A to D Converters:** If there two A to D converters, select either the high speed A to D converter for fast readout (1 MHz for example) or the slower high resolution converter (100 kHz for example) for pixel readout.
- **Gain Index:** Select the gain level compatible with your signal. Typically, there are three gain levels: low, mid, and high. With high gain, fewer electrons (for example, one electron) are used to generate an Analog to Digital Unit (ADU); with low gain, more electrons (for example four electrons) are used.
- **Gain Multification:** Refer to the detector manual for details.
- **ICCD Intensifier and Gain:** Refer to the detector manual for details.
- **Enable Anti-Bloom:** If the CCD has the anti-blooming feature, checking this box will activate it.

The screenshot displays the SpectraSense software interface with the 'CCD' tab selected. The interface includes several sections for configuring the detector settings:

- Optics Tab:** Contains a 'CCD Detector' dropdown menu set to '0: Camera1 EEV 400x1340E'. Below it, the 'Detector Name' is 'Camera1', 'Detector Head' is 'EEV 400x1340B', and 'Detector Size' is '1340(20um) x 400(20um) pixels'.
- Head Temperature:** Shows 'Min: -102 Max: 25'. The 'Operating Temperature' is set to '-70' with a 'set' button.
- Analog to Digital Converter:**
 - Radio buttons for 'Port 0: Low Noise' and 'Port 1: High Capacity' (selected). A 'set' button is present.
 - Radio buttons for 'A to D converter #1' (selected) and 'A to D converter #2'. A 'set' button is present.
 - 'Current AtoD Speed' is '100.0 Khz'.
 - 'Gain Index' is set to 'Gain Level 2'.
 - 'Gain Multification' section includes a 'Factor (0 - 4095)' input field, a 'set' button, and an 'On' checkbox.
 - 'ICCD Intensifier' section includes a dropdown menu, a 'Gain (1-255)' input field, and a 'set' button.
 - An 'Enable Anti Bloom' checkbox is at the bottom.
- Head Orientation:**
 - Radio buttons for '0 degrees' (selected) and '180 degrees'.
 - Radio buttons for 'Use Instrument Calibration' and 'Use Fixed Wavelength File' (selected).
- Cosmic Mode:**
 - Radio buttons for 'As a Count' (selected) and 'As a Percent'.
 - 'Threshold' is set to '50'.
 - An 'Drift Adjust' checkbox is present.
- Wavelength File:** A 'Load File' button.
- Location:** A dropdown menu set to 'Docked to Mono1'.
- Front Exit:** A dropdown menu.

Head Orientation: Click on the appropriate radio button to change the head orientation. If your spectra come out backwards, either invert the head or click on this button to invert the icon.

- **Use Instrument Calibration:** Use the calibration for the instrument the CCD detector is attached to (SpectraGraph, Mono1 ...) Mono2's calibration is not used.
- **Use Fixed Wavelength File:** This parameter is for fixed grating instruments that have no electronics in them (for example, the HTS, InSpectrum-150, and RoseMount). The file is either Acton-generated or user-generated with a tool provided by Acton. When this option is selected, the **Load File** button allows you to locate and load the wavelength file.

Cosmic Mode: Cosmic correction is done using the two frame compare method. When a pixel in one frame and the same pixel in the second frame are non-cosmic, they are averaged. If one of them is a non-cosmic and the pixel in the other frame is cosmic, the cosmic pixel is discarded and the non-cosmic pixel data is used.

The criteria used to determine whether a pixel is cosmic or non-cosmic is determined by the "As a Count" or "As a Percent" choice and the Threshold value. "As a Count" indicates pixel by pixel differences greater than X counts are treated as cosms (this is the default). "As a Percent" option indicates pixel by pixel differences greater than X percent are treated as cosms (this choice is good with bright sources). The threshold is the difference (count or percent) that indicates a cosmic. 50 is good for CCD detector that only cools to -30°C, but a number closer to 10 is best for cameras that cool to -80°C. Drift adjust does a background correction on the frame before doing the pixel by pixel comparisons.

Location: Location allows you to specify the monochromator (if there are two) and the exit port to which the CCD detector is mounted.

Detection tab

The Detection tab appears if a SpectraHub, an NCL, or an activated SR-510 (or SR-810 lock-in) is part of your system.

Note: When a Stanford Research 510 lock-in amplifier (SR-510) is part of the system, it must be activated by selecting Configure SR-510 Lock-In from the Select Config menu. Then, in the SR-510 dialog, you must indicate the Com Port, check "Use SR-510 Lock-In", click on OK, and restart SpectraSense before the lock-in icons are displayed and the Detection tab is available.

SpectraHub or NCL

In this tab you define the type of non-CCD detectors that are supplied with the system, the output units, high voltage settings, if appropriate, location, and the gain.

The standard SpectraHub has one detector channel and is compatible with all ARC PMT, Silicon Diode, IR detectors, and preamps with DC current or voltage output as well as photon counting detectors. Multiple SpectraHubs are controlled by connecting the SpectraHubs to the computer USB ports, computer RS-232 or any USB hub in the system. Note that the SpectraHub and all SP-2000 series Acton series monochromators have built-in USB hubs.

The standard NCL comes with two detector channels, a third panel will appear in systems supplied with three detector inputs. Be sure to verify which detector is attached to each labeled input on the NCL back panel.

Channel Parameters:

- Channel:** Select your detector type from the dropdown list. The choices are:
 - None
 - PMT current
 - Silicon diode (Si diode) current –
 - Silicon diode (Si diode) current +/-
 - Preamplifier voltage + for certain IR detectors
 - Preamplifier voltage - for certain IR detectors
 - Photon Counting
- Maximum HV:** Enter the highest safe voltage setting for your specific detector. No values larger than this will be accepted for input from any other screen. Refer to the detector's manufacturer's documentation for the maximum safe operational voltage. The absolute maximum voltage for a PD471 is 1250 volts. A typical maximum operating value is 800V.
- Gain :** Select X1, X2, X5, X50, X200
- Set button:** Applies the change.
- Abort button:** Cancels any change made in the values.

The screenshot shows the 'Detection' tab in the SpectraSense software. It features three identical channel configuration panels for Channel 1, Channel 2, and Channel 3. Each panel includes a dropdown menu for selecting the detector type, a 'Set' button, a 'Maximum HV' input field with a 'Set' button, a 'Default HV' input field with a 'Set' button, a checked checkbox for 'HV Always Turned Off on Exit', and a 'Location' dropdown menu. At the bottom of the interface, there is a dropdown menu labeled 'Disable Readout Filter Controller'.

Location: These fields report the currently selected monochromator and port to which the detector is mounted. If the detector location has changed, you can select the new locations from these dropdown lists or you can move the icon on the system diagram.

High Voltage:

- **Default HV:** Enter the normal working voltage for your detector. The value entered into this field is not applied until it is confirmed with the **Set** button. You can change this value in the survey screen for specific experimental conditions.
- **On/Off:** This check box is used to have the HV come on when the SpectraHub or NCL is turned on. The change to the check box status is not applied until it is confirmed with the **Set** button.
- **HV Always Turned Off on Exit:** All HV is turned off when SpectraSense is exited. These check boxes cannot be unchecked.

Readout:

- **Reading Units:** Select
 - Readout in Raw Counts
 - Readout in Counts per Second
 - Detector Units (Amps, Volts, Counts Sec for Photon Counters)
- **Enable/Disable Readout Filter Controller:** Apply to the NCL and are not visible with SpectraHub. Enable sets up the NCL filter wheel (see drop down boxes on the Optics tab); Disable removes the NCL filter from the software.

SR-510/SR-810

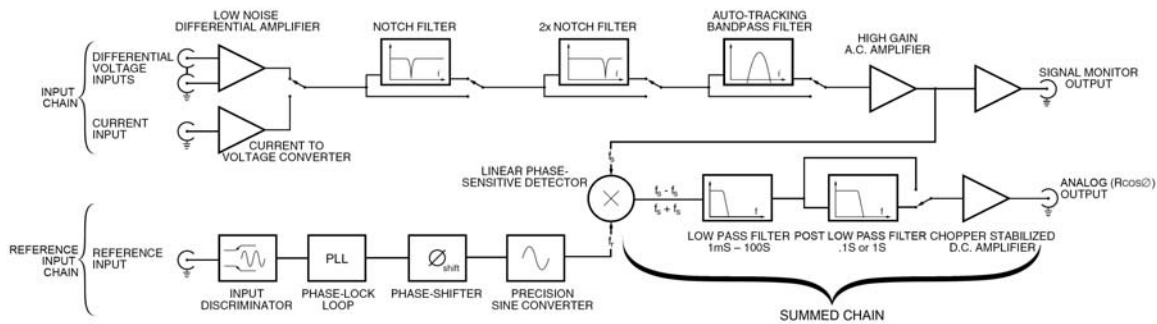


Figure 1. SR-510 Block Diagram

Before Activating an SR-510 or SR-810

SpectraSense supports the Stanford Research SR-510 and SR-810 lock-in amplifiers.

If you will be using one of these instruments, you will need to configure the RS232 communication interface between the computer and the lock-in. This means that you must set the COM port parameters in Windows via **Device Manager** for the computer COM port you will be using and the RS232 settings at the instrument. The actual settings depend on the lock-in you will be using.

Checking and changing the computer's COM port settings may require that you be logged on as an **Administrator**. To open **Device Manager**, click **Start**, point to **Settings**, click **Control Panel**, double-click **System**, click the **Hardware** tab, and then click **Device Manager**.

SR-510 RS232 Communication Configuration

COM Port Settings (via Windows Device Manager): Bits per second: **9600**, Data bits: **8**, Parity: **None**, Stop bits: **2**, Flow control: **None**.

Rear Panel SW2:RS232 mode switch: Bit 1: **Down** and Bits 2-3 **Up** (9600), Bit 5: **Up** (No parity), Bit 6: **Up** (No echo), Bit 7: **Up** (Two stop bits). **The positions of Bits 4 and 8 do not matter.** For more information, please see the SR-510 User Manual.



SR-810 RS232 Communication Configuration

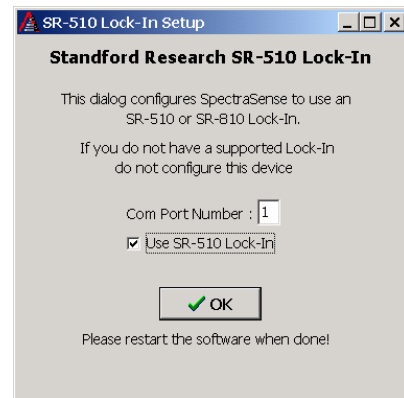
COM Port Settings (via Windows Device Manager): Bits per second: **9600**, Data bits: **8**, Parity: **None**, Stop bits: **1**, Flow control: **None**.

Front Panel Setup Button Settings: RS-232 Baud Rate **9600** and Parity **None**
For more information, please see the SR-810 User Manual.

Before Activating an SR-510 or SR-810

Verify the communication interface is correctly configured for your lock-in amplifier and computer COM port. The SR-510 (and COM port) should be set to **TWO** stop-bits for this model. The SR-810 COM port should be set to **ONE** stop bit. Refer to the previous topic for more information.

The next step is to activate the lock-in amplifier. To do this, select **Configure SR-510 Lock-In** from the **Select Config Menu**. On the SR-510 Lock-In Setup dialog, enter the Com Port number for the device and check in the USE SR-510 Lock-In check box. Then, click on OK. You must now restart SpectraSense before the lock-in icons or the Detection tab for the lock-in amplifier will be available.



Note: To de-activate the lock-in amplifier, you would re-enter this dialog, de-select the check box, click on OK, and then restart SpectraSense.

Setting up an SR-510 or SR-810

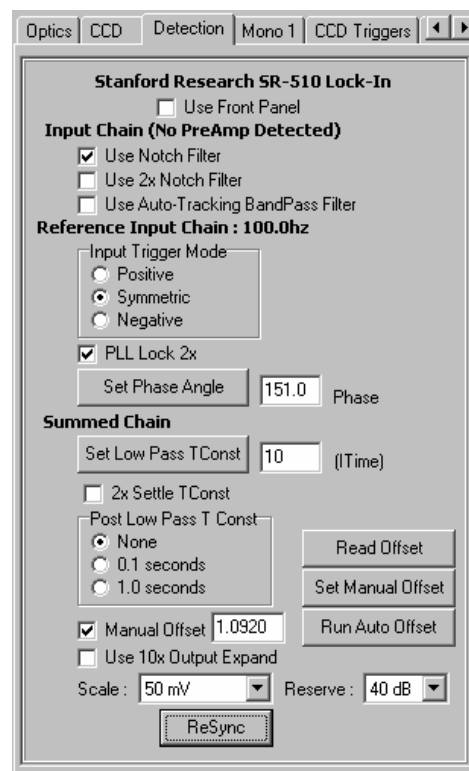
After an SR-510 has been activated, you can begin setting up the signal chain parameters (Figure 1 illustrates the Input, Reference Input, and Summed Input chains). The information in the following paragraphs briefly describes the parameters. For in-depth information, refer to the Stanford Research manual supplied with the lock-in amplifier.

*Note: If a SpectraHub has been configured for the existing system, you will need to select SR-510/SR-810 from the **Optics** tab before the SR-510/SR-810 setup choices will be displayed on the **Detection** tab.*

Use Front Panel: Locks out the software controls and requires that you use the lock-in amplifier's front panel.

Input Chain (No PreAmp Detected): The parameters allow you to choose which filters, if any, that you want in the input chain.

- **Use Notch Filter:** Q of 10 and a depth of at least 50 dB. The line frequency notch is 6 Hz wide.
- **Use 2x Notch Filter:** Q of 10 and a depth of at least 50 dB. The 2x notch is 12 Hz wide.
- **Use Auto-Tracking BandPass Filter:** Q of 5 and a 6 dB rolloff in either direction. The pass band (between 70% pass points) is always equal to 1/5th of the center frequency. The center frequency is continually adjusted to be equal to the internal demodulator frequency. When **PLL Lock 2x** is not active, the filter tracks the reference. When **PLL Lock 2x** is active, the filter frequency is twice the reference input frequency.



Reference Input Chain: These parameters allow you to characterize light chopper signal into the Reference Input BNC on the front of the SR-510. The input is ac coupled and the impedance is 1 M Ω . The dc voltage at this input should not exceed 100 V and the largest ac signal should be less than 10 V peak.

- **Input Trigger Mode:** Defines when the reference oscillator will lock.
- **Positive:** The trigger threshold is +1V and the reference oscillator will lock to the positive-going transitions of the reference input. This mode triggers on the rising edges of a TTL type pulse train. The pulse width must be greater than 1 μ s.

Symmetric: The reference oscillator will lock to the positive zero crossings of the ac reference input. The ac signal must be symmetric (e.g., sine wave, square wave, etc.) and have a peak to peak amplitude greater than 100 mV. A signal with 1 Vrms amplitude is recommended. The phase accuracy of the reference channel is specified for a 1Vrms sine wave in the symmetric trigger mode.

Negative: The trigger threshold is -1V and the reference oscillator will lock to the negative-going transitions of the reference input. This mode triggers on a negative pulse train or on the falling edges of a TTL type pulse train (remembering that the input is ac coupled). The pulse width must be greater than 1 μ s.

- **PLL Lock 2x:** When selected, the PLL will lock to twice the reference input frequency. In this mode, the reference input frequency may not exceed 50 kHz. When this mode is not selected the reference input frequency may not exceed 100 kHz.
- **Set Phase Angle:** Allows you to enter and then set the phase shift between the reference oscillator and the reference input.

Summed Chain: The Summed Chain components filter and offset the output signal before it reaches the **Output BNC**. There are two post demodulator low pass filters, Low Pass and Post Low Pass. The Low Pass filter precedes the Post Low Pass filter in the output signal chain. Each filter provides 6 dB/oct attenuation. The offsets (either manual offset or those generated by the **auto offset** function) represent a fraction of the full scale reading, and so their absolute value will change when the Gain is changed. A signal which has been nulled by an offset will not be nulled when the Gain is changed.

- **Set Low Pass Time Constant:** Enter the time constant. The range is from 1 ms to 100 s.
- **2x Settle T Const:** When checked, the system waits twice the Time Constant between reads instead of the Time Constant as entered above. This allows additional time for the signal to reach a steady value before a reading is taken.
- **Post Low Pass T Const:** Can be set to None, 0.1 second, or 1.0 second. If set to None, the Post Low Pass filter will be not be included in the chain and the total attenuation is 6 dB/oct. When 1.0 second or 0.1 second is entered, the Post Low Pass filter will be

Optics CCD Detection Mono 1 CCD Triggers

Stanford Research SR-510 Lock-In

☐ Use Front Panel

Input Chain (No PreAmp Detected)

☒ Use Notch Filter

☐ Use 2x Notch Filter

☐ Use Auto-Tracking BandPass Filter

Reference Input Chain : 100.0Hz

Input Trigger Mode

☐ Positive

☒ Symmetric

☐ Negative

☒ PLL Lock 2x

Set Phase Angle 151.0 Phase

Summed Chain

Set Low Pass TConst 10 (ITime)

☐ 2x Settle TConst

Post Low Pass T Const

☒ None

☐ 0.1 seconds

☐ 1.0 seconds

☒ Manual Offset 1.0920

☐ Use 10x Output Expand

Scale : 50 mV Reserve : 40 dB

Read Offset

Set Manual Offset

Run Auto Offset

ReSync

included and the total attenuation is 12 dB/oct for frequency components beyond the larger of the two filter bandwidths (reciprocal time constant).

- **Read Offset:** Reads the current offset and reports it in the field to the right of **Manual Offset**.
- **Manual Offset and Set Manual Offset:** When the **Manual Offset** box is checked, auto offset is turned off and the offset entered in the field to the right of Manual Offset will be applied to the output signal. If Auto Offset was run and then Manual Offset is activated, the offset will be that set by the **Auto Offset** function. To change the offset, enter the new value in the field and click on the **Set Manual Offset** button.
- **Run Auto Offset:** Turns off manual offset and runs the auto offset function for injection in the summed chain. The offset value is set to minus the value of the available at the **OUTPUT BNC**, thus zeroing that output. If the output is greater than 1.024 times full scale, the **Auto Offset** function will not be able to zero the output and the offset value will then be set to its maximum value.

Note: The Auto Offset function only works on the SR-510: it does not work on SR compatibles.

- **Use 10x Output Expand:** Multiply the output signal by 10.
- **Scale:** The SR-510 has full scale sensitivity from 100 nV to 500 mV RMS for voltage inputs or from 100 fA to 500 nA RMS for current inputs. The SR-810 has full scale sensitivity from 2 nV to 1 V RMS for voltage inputs or from 2 fA to 1 μ A RMS for current inputs. The sensitivity indication is not changed by the **10x Output Expand** function. The SR-510 and SR-810 each has a current input; other SR-compatible lock-in amplifiers may not have one. While the SR-810 can be set to 1 V, this setting is changed to .5 V when the setup is saved.
- **Reserve:** Select the dynamic reserve appropriate for the sensitivity.
 - 20 dB LOW (1 μ V to 500 mV sensitivity)
 - 40 dB NORM (100 nV to 50 mV sensitivity)
 - 60 dB HIGH (100 nV to 5 mV sensitivity)

Note: The Bandpass filter adds up to 20 dB to dynamic reserve for noise signals outside the pass band and increases the harmonic rejection by at least 13dB. The Notch filters increase dynamic reserve to 100 dB. If the scale is changed to a setting for which the dynamic reserve is not allowed, the dynamic reserve will change to the next setting which is allowed.

ReSync: Updates the Detection tab entries with the settings from the front panel of the SR-510.

Saving/Reloading the Detection Settings

You can save the configuration you are using for the lock-in amplifier on the **Acquisition** tab. Select *Single Point Scan* from the *Scan Type* dropdown list.

The screenshot shows the SpectraSense software window with the 'Acquisition' tab selected. The 'Scan Type' dropdown menu is open, showing options: 'Single Point Scan Mono1', 'Single Point Scan Mono2', 'Single Point Intensity vs Time', and 'CCD Spectral Acquire'. The 'From' field is set to 1.0, 'To' is 10.000 nm, and 'Step' is 1.000 nm. Channel 2 is set to 0 and Channel 3 is set to 3210. The 'Realtime Processing' section shows 'Channel' set to 'Ch 1', 'ITime' set to 100, 'Reads per point' set to 1, 'Cycles' set to 1, and 'Delay' set to 0 h 0 m 0 s. The 'Filter 1' is set to 0. The 'Storage' section shows 'Single File per Scan' and 'Format' set to 'Incremental (postfix)'. The 'Acquisition Routine' section shows 'Routine Not Saved !!!' and 'Change' button. The 'Routine' section has 'Save', 'Save As', and 'View' buttons. The 'Settings' checkbox is checked, and 'Match Hardware' is also checked. The 'Operator' field is empty, and the 'Comments' field is empty. The 'Save data as' field is empty, and the 'Export To Microsoft Excel' checkbox is unchecked. The 'Acquire' and 'Cancel' buttons are at the bottom.

If you “**Save As**”, you can create and save a file containing the software settings you have set on **Detection** tab and scan settings (if entered) for the lock-in amplifier. You can then reload this file either from the **Acquisition Routine** entry (same session) or via the “**Change**” button.

If you select the **Match Hardware** checkbox before reloading the file, SpectraSense will set the lock-in amplifier with the saved settings when the file is reloaded. If this checkbox is not selected, only the settings in SpectraSense independent of the hardware will be restored.

Notes:

1. If the SR-510 appears to be working, but does not capture data or certain settings, verify that the COM Port is set to **TWO** stop bits and that Bit 7 on the SW2 RS232 switch on the back the lock-in is in the **UP** position.
2. If you are using an SR-810 and **Scale** is set to **1 V**, this setting will be saved as **.5 V** due to the limitations of SR-510 compatibility. If you later restore this SR-810 setup, you will have to re-set **Scale** to **1 V**.

Monochromator tab

The monochromator tab is for the most part informational. All of the monochromator characteristics including grating types and port configurations are read directly from a flash ram in the monochromator. Although it is possible to change the specification of a port, for field upgrade, it is recommended that these fields not be modified by the user.

The model number and serial number of the monochromator are automatically entered into the specified fields.

Turrets and Gratings: Your monochromator may be supplied with interchangeable turrets. If that is the case gratings 1 through 3 are installed on turret number 1, gratings 4 through 6 on turret number 2 etc. Turrets must be manually changed and the corresponding turret number selected in the turret field. The grating field will toggle through the available gratings on the specified turret showing their groove density, blaze wavelength, and wavelength at maximum rotation.

Double Offset: (Mono2 tab) This only appears for two-stage monochromators. It is the wavelength offset of the slave from the master. For example, if the master is instructed to go to 546.074 nm and the offset is 1.0 nm, the master will go to 546.074 nm and the slave will go to 547.074 nm.

Slits: To activate a manual or motorized slit, click on **Set Active** button for that slit. If your monochromator is supplied with motorized slits, you can set a specific width in microns by entering a value in the field and clicking on the **set** button. You can recalibrate the slit setting by clicking on the **home** button. The band pass for that specific slit width with the specified grating at the current central wavelength is shown below.

If the **Use BandPass Slits** box is checked, then the values entered into the fields should be the band pass in the selected units. The width in microns will be shown below. Selecting constant band pass slits is only appropriate with motorized slits as the software must adjust the widths during operation.

Optics | CCD | Mono 1 | Mono 2 | CCD Triggers | F | ▶

MonoChromator 1

Model # SP-2-500i
Serial # 25580320

Turret : 1
Grating 1
3600 g/mm
Blaze = H-UV
467 nm cutoff

1	3600 g/mm, Blaze = H-UV
2	2400 g/mm, Blaze = H-VIS
3	600 g/mm, Blaze = 500nm
4	Not Installed
5	Not Installed
6	Not Installed
7	Not Installed
8	Not Installed
9	Not Installed

Side Exit Slit Set Active
Manual Slit

Side Entry Slit Set Active
Motorized Slit
50 set home
BandPass : 0.027 nm

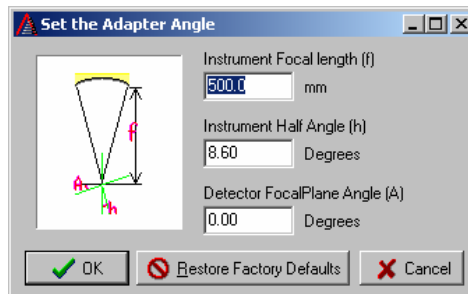
Front Exit Slit **Active**
Focalplane Adapter

Front Entry Slit **Active**
Manual Slit

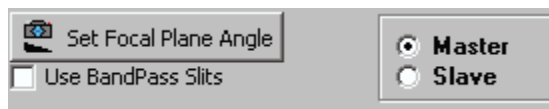
Set Focal Plane Angle
☐ Use BandPass Slits

Note: InSpectrum does not support constant band pass operation. However, the band pass for each width is displayed.

Set Focal Plane Angle button: Used when an angled CCD adapter flange is supplied with the spectrograph. The default values are the specified focal length of the instrument and 0 degrees for the angle. If an angled adapter is supplied with the spectrograph the value should be changed to 11 degrees.



Master/Slave: Associated with double monochromator (two monochromators on a single base with only one USB or RS-232 connection to the computer). These radio buttons allow you to toggle between the two stages. They are not visible for single-stage monochromators.



Filter Wheel tab

If an automated filter wheel has been detected in your system, the filter wheel tab will appear. The filter wheel holds up to six filters except in the case of *InSpectrum* where there are only five positions. The wheel can be set up to automatically insert cut off filters at specified wavelengths and to allow insertion of a specific filter to remain in place for the entire acquisition.

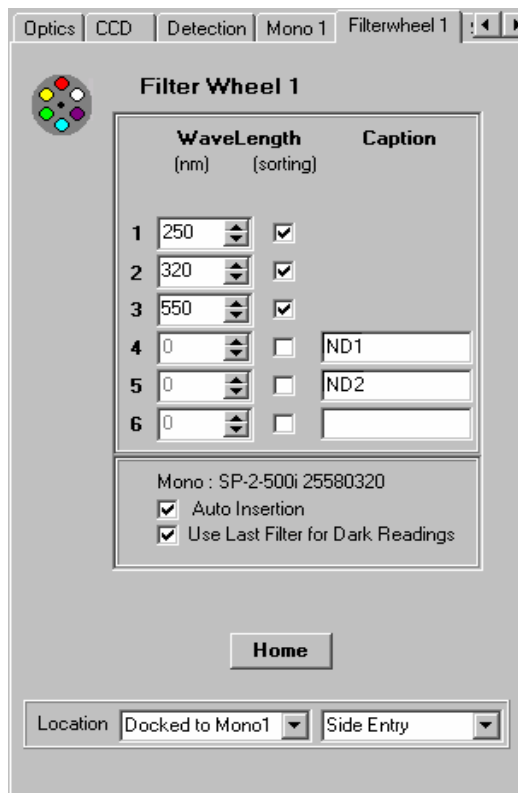
Wavelength Sorting: If a sorting filter is installed in a specific position, enter the wavelength in nm that it is to be inserted and check the sorting box. Other types of filters such as neutral density filters can be installed in the wheel. For these types of filters, the sorting box is left unchecked and a caption box is opened for naming the filter. Wavelength sorting is for the monochromator to which the filter wheel is assigned.

Auto Insertion: In a scanning acquisition, the monochromator will momentarily stop while the appropriate filter is inserted if the **Auto Insertion** box is checked.

Dark Readings: An opaque filter can be inserted in the last position to be used as a shutter for dark subtraction. If this is the case, check the **Use Last Filter for Dark readings** check box.

HOME: The **HOME** button will reset the wheel in the case where the knob was used to move the wheel.

Location: These fields report where the filter wheel is currently positioned on the system diagram. If the filter wheel location has changed, select the monochromator and entry port to which the wheel is actually mounted.



Shutters tab

This tab deals with non-CCD blocking shutters and appears if shutters were specified in the Optics tab.

Note: There are two classes of shutters: CCD and blocking. The CCD shutter is intimately integrated with the readout of the CCD. Non-CCD shutters are used for dark subtraction and PMT protection in scanning operation.

Shutter: A separate panel is created for each shutter activated on the Optics tab. Note that only Shutter 1 has the CCD Shutter check box.

CCD Shutter: Click on this check box if the shutter is to be associated with a CCD. It may also be associated with an entrance slit for scanning operation.

Shutter Safety Zone: The shutter safety zone is used to protect sensitive detectors from overexposure to sources such as lasers. Enter the wavelength to be protected from in the center field. Enter the number of working units, nm, cm⁻¹ as specified in the Optics tab, on either side of the center wavelength over which the shutter will remain closed. For example, for a Raman experiment 0 relative wavenumbers would be specified as the center with a safety zone of +/- 20 wavenumbers. The shutter would remain closed between -20 and 20 cm⁻¹.

Enable Safety Zone: This check box activates/deactivates the shutter safety zone feature.

Location: These fields report the currently selected monochromator and port to which the shutter is mounted. If the shutter location has changed, you can select the new location from these dropdown lists or you can move the icon on the system diagram.

The screenshot displays the 'Shutters' tab within a software interface. At the top, a navigation bar includes tabs for 'Detection', 'Mono 1', 'Filterwheel 1', 'Shutters', and 'NCL Tr'. The 'Shutters' tab is active. Below the navigation bar, there are two main sections for 'Shutter 1' and 'Shutter 2'. Each section contains a shutter icon, a 'CCD Shutter' checkbox (checked for Shutter 1, unchecked for Shutter 2), and a 'Shutter Safety Zone' configuration area. The 'Shutter Safety Zone' area includes two input fields for 'Wavelength (nm)' labeled 'Center' and 'Safety +/-', and an 'Enable Safety Zone' checkbox (checked for Shutter 1, unchecked for Shutter 2). Below these settings, there are 'Location' dropdown menus. For Shutter 1, the location is 'Docked to Mono1' and 'Side Entry'. For Shutter 2, the location is 'UnDocked'.

NCL Triggers tab

SpectraHub

If your system uses a SpectraHub, the NCL triggers tab reports the input trigger status and allows you to control the shutter control for the SpectraHub.

Input Triggers: Reports status of Input Trigger(s). If an input trigger is on, SpectraSense will wait for a hardware trigger on the trigger input line before taking a reading. In the illustration to the left, the Input triggers are ON. To turn them off, click on the circle .

Shutter Control: Allows you to enable or disable the shutter and to select the TTL level that will open the shutter.

NCL

If your system uses an NCL, this tab is used to specify triggering conditions for the input and output trigger lines included in the NCL. It is your responsibility to determine the type of trigger output is needed to communicate with ancillary devices. Refer to the NCL manual for pin assignments for each trigger.

Input Triggers: The four input triggers in the NCL can be used to initiate the following actions:

- Inactive (default setting)
- Start an acquisition routine
- Start a cycle within an acquisition routine
- Acquire the next data point
- End the acquisition (Abort)

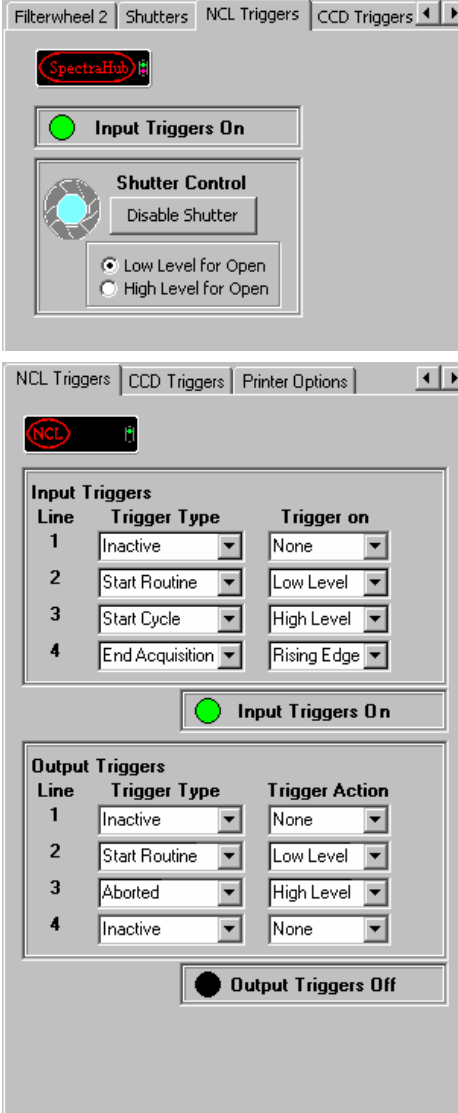
These triggers may be of the following type as provided by the triggering device:

- None
- Low level
- High level

In the illustration, the Input triggers are ON. To turn them off, click on the circle .

Output Triggers: The NCL is also capable of outputting TTL triggers to ancillary equipment. The Trigger Action options are **None**, **Low level**, and **High** level. Triggers may be initiated by the following conditions:

- Inactive (default setting)
- Start of acquisition routine
- Start of acquisition cycle
- End of acquisition cycle
- End of acquisition routine
- Abortion of routine or cycle



Filterwheel 2 | Shutters | NCL Triggers | CCD Triggers

SpectraHub

☒ Input Triggers On

Shutter Control

Disable Shutter

☒ Low Level for Open
☐ High Level for Open

NCL Triggers | CCD Triggers | Printer Options

NCL

Input Triggers

Line	Trigger Type	Trigger on
1	Inactive	None
2	Start Routine	Low Level
3	Start Cycle	High Level
4	End Acquisition	Rising Edge

☒ Input Triggers On

Output Triggers

Line	Trigger Type	Trigger Action
1	Inactive	None
2	Start Routine	Low Level
3	Aborted	High Level
4	Inactive	None

☐ Output Triggers Off

CCD Triggers tab

The CCD triggers tab will appear only if there is a CCD camera detected by the software. Whenever there is a Help topic about a parameter, you can access that help by clicking on the associated  button.

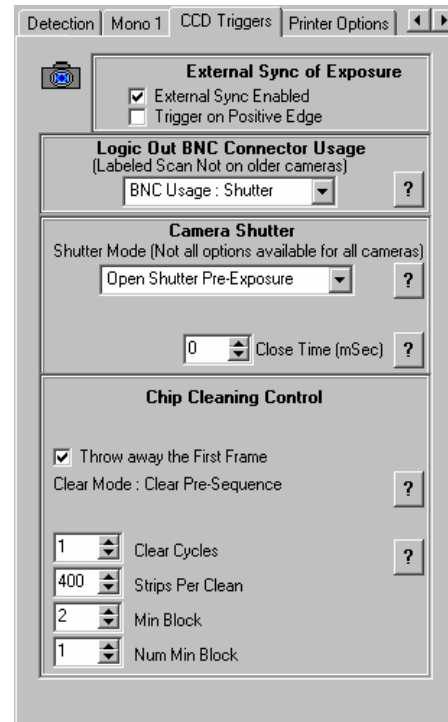
External Sync of Exposure: Check the **External Sync Enabled** box to synchronize the readout of the CCD to an external signal provided by another device. The negative edge will trigger the exposure unless you check in the **Trigger on Positive Edge** box. Refer to the detector or system manual for specifications of the type of triggers accepted.

Logic Out BNC: Choices for this output depend on the detector. Among the choices are Not Scan, Shutter and Not Ready. For more information about the logic out conditions, click on the  button or refer to the detector or system manual.

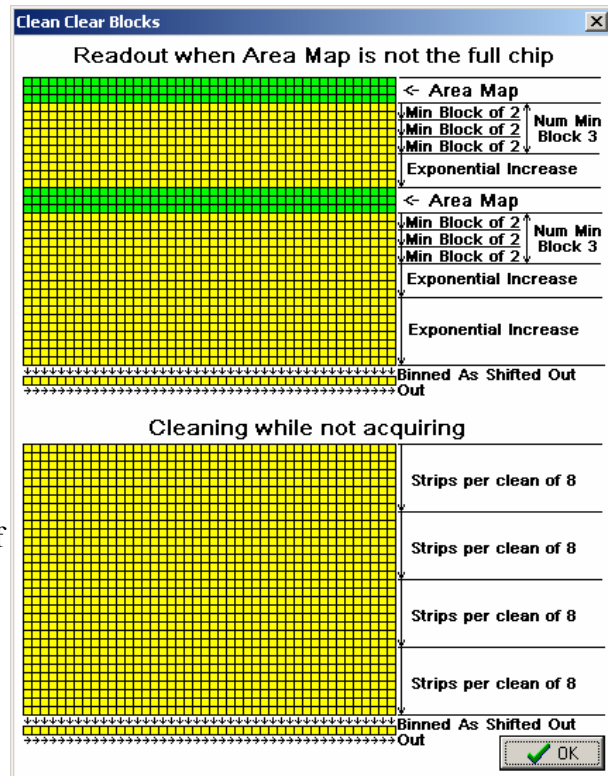
Camera Shutter : The shutter is normally closed when there is no acquisition in progress. Typically, the shutter mode would be **Open Shutter Pre-Exposure** (the shutter opens at each exposure). However, other modes may be available. However, other modes may be available. The **Open Time** and **Close Time** settings only appear when the detector supports them. These settings are delays before and after each readout to compensate for the time it takes the shutter to open and close. Typical values for an *InSpectrum* are 35 ms for open and 8 ms for close. For more information a  at the shutter modes, click on the  button or refer to the detector or system manual.

Chip Cleaning Control: When the CCD is not being read out, the detector is still acquiring dark charge. This can accumulate to a significant value if there are many seconds between acquisitions.

- **Throw away the First Frame:** When an external source is being used to initiate a multiple readout acquisition, you may want to throw away the first readout. Since the amount of time from initialization of the experiment to the first External Sync pulse is not fixed, an accurate background subtraction may not be possible for the first readout. In multiple readout acquisition experiments, this is easily overcome by simply discarding the first frame.
- **Clear Mode:** The selected mode determines how the CCD will be kept clean between acquisitions to prevent the accumulation of acquired dark signal. In many cases, you may have no choice. However, if you do, click on the  button or refer to the detector or system manual for more information.



- Clean/Clear Blocks:** The **Clear Cycles** and **Strips per Clean** parameters allow you to program additional clean cycles that will be performed after initiating the run but before any data is taken. **Clear Cycles** is also used to define the number of clear cycles for the selected Clear Mode. The **Min Block** parameter sets the size, in strips, of the skip blocks that immediately precede the data. The default value will generally give good results. The **Num Min Block** parameter sets the number of binned "skip" blocks preceding and following the region of interest. The default value will generally give good results.



Printer Options tab

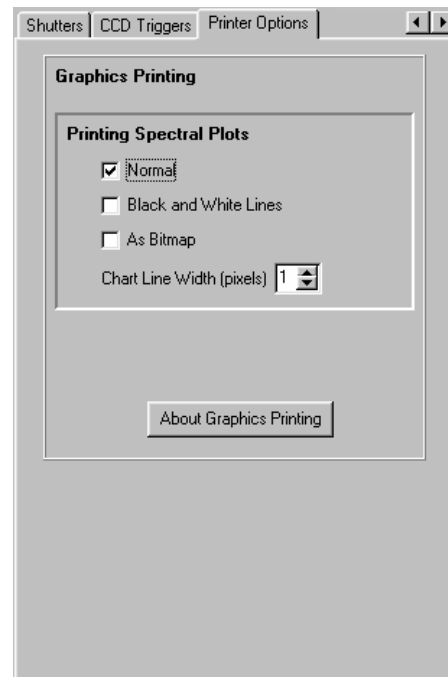
The Printer options tab is used to correct for known problems with several printers and to switch between back and white and color printing.

Printing Spectral Plots:

- Normal:** Used as your default option if your printer is 100% compatible with Windows 98/NT.
- Black and White Lines:** Used if your laser printer has problems converting the color lines into gray scale.
- As Bitmap:** Used as a last resort if your printer is not responding correctly.

Chart Line Width: Used to make the axis and plots thicker. Some printers have a problem printing one pixel width.

About Graphics Printing: Pops up a dialog box that explains why some printers may not correctly handle graphics.



Menu Options

File

Exit	Exits the software.
------	---------------------

Select Tab

Hardware Config	Calls up the Hardware Config screen.
Slit Band Pass	Calls up the Monochromator Slit BandPass screen.
Survey Mode	Calls up the Survey mode.
Acquisition	Calls up the Acquisition screen.
Live Data	Calls up the Live Data screen.
Post Processing	Calls up the Post Processing screen.
Mono Control Software	Calls up the Acton Monochromator Control Software screen.
Terminal	Calls up terminal mode for communications debugging. Used for diagnostics in communications between the hardware and computer.

Select Config

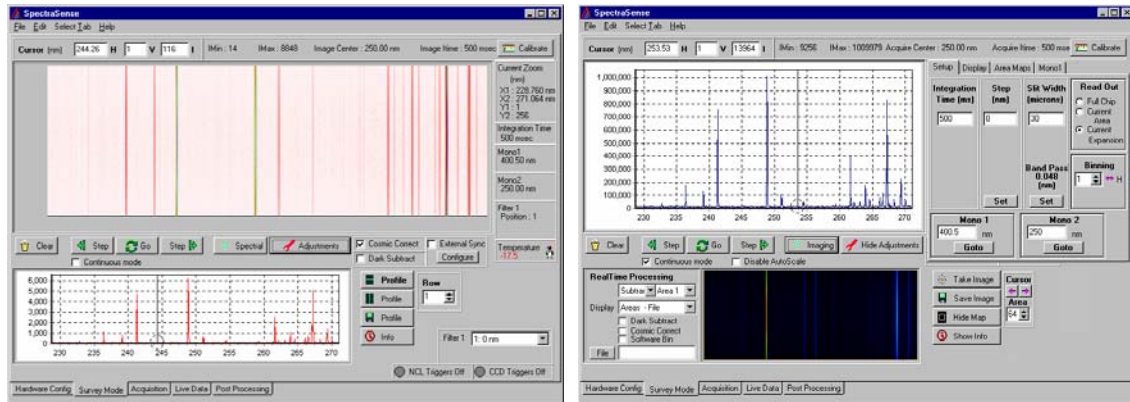
Optics Setup	Jumps to Optics tab on panel.
CCD Setup	Jumps to CCD tab on panel if a CCD is detected.
Detectors Setup	Jumps to Detectors tab if an NCL is detected.
Mono 1 Setup	Jumps to Mono 1 tab.
Mono 2 Setup	Jumps to Mono 2 tab if a second monochromator is detected.
Filter Wheel Setup	Jumps to Filter wheel tab if one is specified in the Optics tab.
Shutter(s) Setup	Jumps to Shutters tab if shutters are specified in the Optics tab.
NCL Triggers Setup	Jumps to NCL triggers tab if NCL is detected.
CCD Triggers Setup	Jumps to CCD triggers tab if a CCD is detected.
Printer Setup	Jumps to Printer tab.
Configure SR-510 Lock-In	Opens the SR-510 Lock-In Setup dialog where you can specify the Com port and activate or de-activate the lock-in. Note, after activating the lock-in on this dialog, you must restart SpectraSense before the icons or the Detection tab for the lock-in will be available.

Help

Contents	Hyperlinked index of the SpectraSense manual
Overview of SpectraSense	Complete SpectraSense manual on line
Using Hardware Config	Hyperlinked access to this chapter
SpectraSense Version	Version number of the installed software

This page intentionally left blank.

Chapter 3: CCD Survey Screen



CCD Survey Mode

The CCD survey mode screen appears when the CCD detector is in the optical path and the Survey screen tab is selected either from the menu level or from the tab at the bottom of the page.

The purpose of the survey screen is to optimize the acquisition parameters for taking spectra. From within this screen you can adjust the spectrograph position, the integration time, area(s) of the chip from where the spectra will be acquired and slit widths. These parameters control the signal to ratio of the spectra and the spectral range that will be covered.

You can adjust these parameters in both a spectral mode and an imaging mode. The combination of both modes provides an accurate and precise method for optimizing spectral acquisition. The top window of this screen shows the active mode. The bottom window is used to help evaluate the active mode. Spectra and images can be directly stored from within this screen, however the main purpose is to transfer the operating parameters to the acquisition screen which provides more flexible data acquisition and storage of routines that can be called upon repeatedly. The spectral and image displays in both windows can be expanded or returned to full scale using the mouse dragging procedure discussed earlier in this document.

Main Display Window and Functions

Acquisition Mode and Adjustments buttons

Acquisition Mode: You select your acquisition mode by clicking on the two-state Imaging/Spectral button. The present state of the button indicates what mode will be active if the button is clicked on. In the illustrations above, the imaging mode is active on the left and spectral on the right.



Adjustments: The two-state Adjustments button is used to toggle between a larger primary display and access to the panels that control the acquisition and viewing parameters. In the illustrations above, the adjustment panels are hidden on the left and visible on the right.



Cursor Coordinates panel

The cursor coordinates panel (located above the primary display window) shows the horizontal position of the cursor in the selected working units. In imaging mode the second field, V, displays the column number of the cursor. In spectral mode it displays from which area the intensity value is being displayed in the intensity field. In imaging mode the intensity field shows the intensity at the specific pixel with the displayed H and V coordinates.

Cursor (nm) 246.34 H 4 V 46 I

Go/Stop: The Go/Stop two-state button is used to acquire data.

If the **Continuous** box is checked, data will be continually acquired and the display updated until the Stop button is pressed. The Step buttons are used to move the spectrograph a specified number of wavelength units to either higher or lower wavelength before each read acquisition. If the continuous box is checked the spectrograph will continuously move to either high or lower wavelength and update the display. The increment for the motion is specified in the **Setup** tab.

Step Go Step

Clear: The **Clear** button erases the image or spectrum displayed in both windows.

Clear

Adjustments tabs

Setup tab

The Setup tab controls most of the parameters that affect the spectral quality and wavelength coverage.

Integration Panel: The integration time is how long the CCD will be exposed before the data is read out. Doubling the integration time will double the signal intensity.

Step Panel: The Step panel is used to define the incremental position change for each click of the left or right step arrows. After entering a step increment in your selected spectral units, click on the Set button to validate the change.

Slit Width Panel: The Slit width panel is used to set the width of the slit (if motorized). The spectral band pass for the width is shown on the panel. (If you have selected band pass units for the adjustments on the Hardware Config screen, the field will show the band pass and the information below will show how wide the slit is in microns. You must validate your entry for automated slits before the change will occur by clicking on the **Set** button.

Read Out Panel: The Read Out panel defines what portion of the chip will be displayed in image mode only. If Full Chip is selected the entire chip will be displayed. If Current Area is selected, only the defined areas of the chip will displayed. If Current Expansion is selected, only the current visible expanded portion of the chip will be updated and displayed. The smaller the portion of the chip you select, the faster the image will update. For focusing fiber optics for example it would wise to expand the image to show only a few of the fibers. In this way you can iteratively adjust the focus more quickly.

Setup | Display | Area Maps | Mono1 | Mono2

Integration Time (ms) 500

Step (nm) 0 **Set**

Slit Width (microns) 30 **Set**

Band Pass 0.048 (nm)

Read Out

- ☐ Full Chip
- ☐ Current Area
- ☒ Current Expansion

Binning 1 **H**

Mono 1 346.5 nm **Goto**

Mono 2 400.5 nm **Goto**

Binning Panel: The binning panel will have both H(orizontal) and V(ertical) adjustments in imaging mode and only horizontal in spectral mode. By binning pixels together you can increase the signal to noise at the expense of spectral or image resolution. In many cases binning by two pixels or more will not degrade the spectral resolution and will increase the signal to noise ratio.

Mono Panels: The Mono panels show the current central wavelength of the spectrograph(s). You can move the spectrograph by changing the value in the position field and clicking on the **Goto** button.

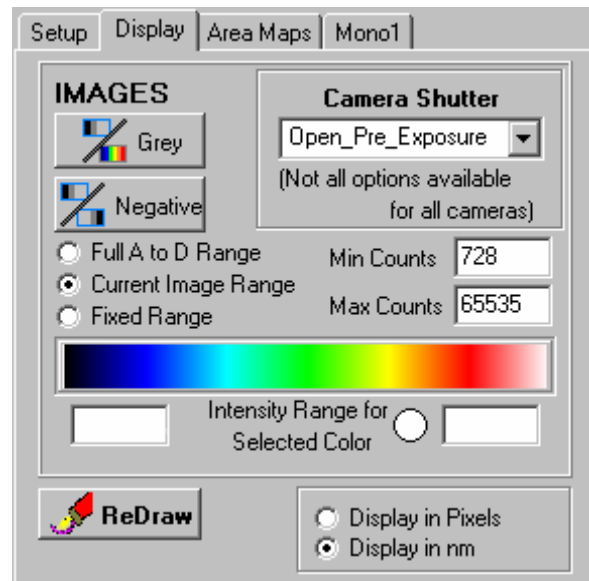
Display tab

The **Display tab** is primarily used to enhance the images displayed in the top window. It is also here that the x axis of the spectrum and image can be shown in wavelength units or horizontal pixel position.

Grey/Color button: Changes the image between false color and gray scale.

Negative/Positive button : Reverses the color scale. Sometimes weak features are best visualized in negative grey scale.

Camera Shutter: Allows you to set the shutter mode. The shutter is normally closed when there is no acquisition in progress. Typically, the shutter mode would be **Open Shutter Pre-Exposure** (the shutter opens at each exposure).



Color Range: To find the count range for any color in the image, click on the same color on the color band in the panel. The minimum and maximum counts for that color will appear in the intensity range fields.

Full A to D Range Displays the full level of colors between 0 and 65,535 counts for a 16 bit ADC or between 0 and 4096 counts for a 12 bit ADC, irrespective of the actual count range for the image.

Current Image Range: Sets the color levels from the minimum intensity value to the maximum intensity value in the current image.

Fixed Range button: Allows you to set arbitrary minimum and maximum intensities over which the color levels will be displayed.

If you have changed any of the parameters discussed above, you can redraw the image with the new settings by clicking on the **ReDraw** button.

Display in Pixels/Display in nm: You must re-acquire the image or spectrum after changing the display between pixels and spectral units.

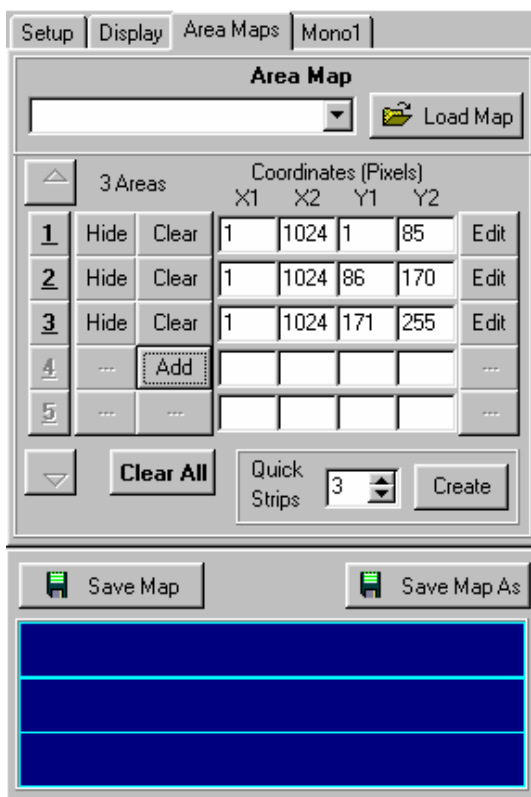
Area Maps tab

This tab is used to define the areas on the CCD chip from which spectral or image data will be acquired. Multiple areas can be defined as long as there is no overlap between areas. Areas are important in that the size of an area can have a significant effect on the signal to noise ratio of a spectrum and in reducing saturation in cases where there is a strong signal. In working with fiber optics, you would normally define areas that just cover the height of each fiber and the width of the chip.

There are three ways in which areas can be defined. However before any new area maps can be defined you should first click on the **Clear All** button and then on the **Add** button.

1. You can draw areas on an image that is displayed in the top window by clicking and dragging the mouse to draw a box around the area,
2. You can enter x and y limits, in pixel positions into the coordinates fields, or
3. You can use the **Quick Strips** function to create up to 32 equal height areas across the chip. You can clear or add areas by clicking on the clear/add button. You can edit the values in the coordinates fields by clicking on the Edit/Set button. You can choose to display or hide any of the areas by clicking on the Hide/Show button. When using the Quick Strips method of creating areas, sometimes there will be a group of pixels left over that is smaller than the others. These will show up as another area. You may choose to clear this area or hide it from display.

As you define the areas they will be displayed in the diagram at the bottom of the panel. After you have created a map of the desired areas, you can save it for future use by clicking on the Save Map button. You can reload the map at any time by clicking on the Load Map button. If you modify a saved map and want to keep both the original and modified versions use the **Save Map As** button to save the modified version.



Mono1 tab

The Mono1 tab is associated with the first monochromator in the system. It allows you to change the width of a motorized slit, set the step size, change the grating, go to a specific wavelength, or step as specified by the step size.

Slit Widths: You can adjust the slit widths of motorized slits in the Mono1 light path by entering the desired values in the fields and clicking on the **Set** button. If **Use BandPass Slits** was selected on the Hardware Config screen and the slit is motorized, you can enter the band pass value. Otherwise, it is only reported for motorized slits.

Step (nm): The monochromator can be set to step the number of spectral units entered.

Mono: The position of the monochromator is visible in the **Mono** panel and can be changed by entering a new position and clicking on the **Goto** button.

Grating Selection: If there are multiple gratings, you can choose the desired grating from the drop down list.

Step buttons: Allow you to step the monochromator to the next position.

Mono2 tab

If there is a second monochromator in your system, a **Mono2 tab** will appear.

Slit Widths: You can adjust the slit widths of motorized slits in the Mono2 light path by entering the desired values in the fields and clicking on the **Set** button. If **Use BandPass Slits** was selected on the Hardware Config screen and the slit is motorized, you can enter the band pass value. Otherwise, it is only reported for motorized slits.

Inter Slit Width appears if you have a double monochromator.

Step (nm): The monochromator can be set to step the number of spectral units entered.

Mono: The position of the monochromator is visible in the **Mono** panel and can be changed by entering a new position and clicking on the **Goto** button.

Grating Selection: If there are multiple gratings, you can choose the desired grating from the drop down list.

Step with Reads: If the **Step with Reads** box is checked, the monochromator will move to the next position each time the CCD is read out. This is useful for acquiring fluorescence excitation emission matrices.

Step buttons: Allow you to manually step the monochromator to the next position.

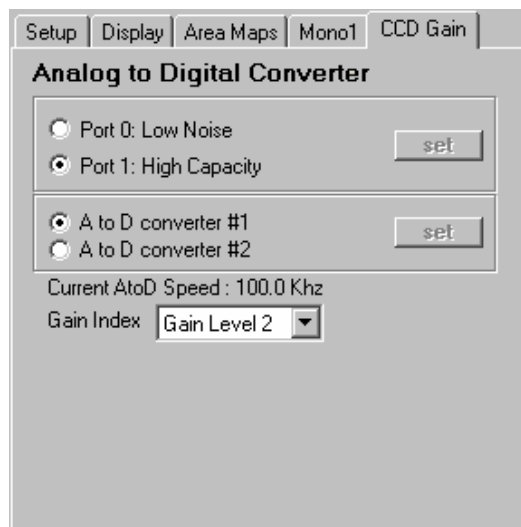
CCD Gain tab

The CCD Gain tab appears if the CCD detector has multiple ports, multiple Analog to Digital (A to D) converters, and/or selectable gain indices.

Ports: If there are two readout ports, select the one that meets your needs: low noise if your signal has a small signal to noise (SNR) ratio or high capacity if there is a high SNR. If you change the current setting, click on the **Set** button to apply the change.

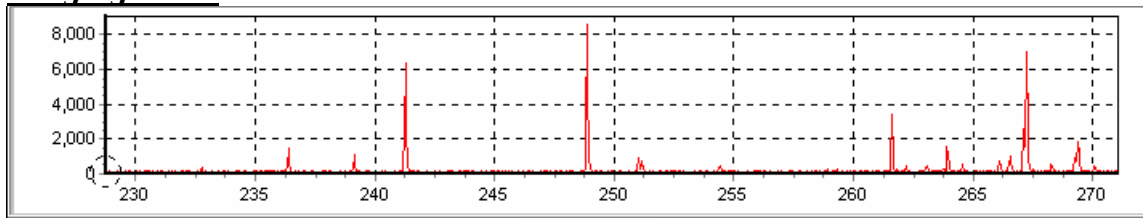
A to D Converters: If there two A to D converters, select either the high speed A to D converter for fast readout (1 MHz for example) or the slower high resolution converter (100 kHz for example) for pixel readout. If you change the current setting, click on the **Set** button to apply the change.

Gain Index: Select the gain level compatible with your signal. Typically, there are three gain levels: low, mid, and high. With high gain, fewer electrons (for example, one electron) are used to generate an Analog to Digital Unit (ADU); with low gain, more electrons (for example four electrons) are used.

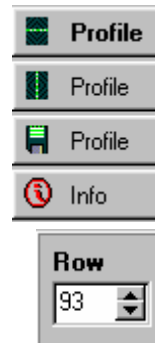


Lower Display Window and Functions

Imaging Mode



If you are in image mode, the smaller window can be used to analyze the image by providing either horizontal or vertical profiles of the image. These show up in graphical form. The **Horizontal Profile** button will display the intensities across a single row of pixels. The row can be specified by either moving the cursor to the desired position on the image or by entering a value in the **Row** panel. A **Vertical Profile** will give a graph of intensity versus height along the entrance slit. You can select the column to be displayed by either moving the cursor to the desired position or entering a value in the column panel. You may save the profile data in a file by clicking on the **Save Profile** button.

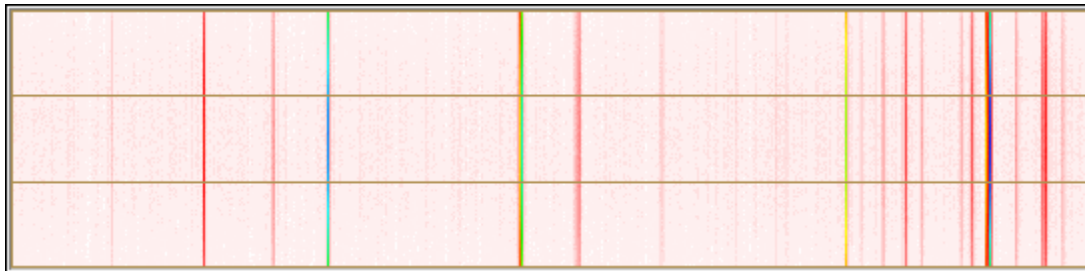


Clicking on the **Info** button will clear the display and show a panel containing all of the acquisition parameters.

Acquisition	Excitation	Area Map
Integration : 500 msec	Position : 400.50 nm	Not Save
Emission	Entry Slit : Side	3 Areas
Position : 250.00 nm	Slit Width : 30 um	3 Active
Entry Slit : Side	Exit Slit : Front	H Bin = 1
Slit Width : 30 um	Slit Width : 10 um	V Bin = 1
Grating : 1	Grating : 1	
Grating : 1200 g/mm	Grating : 600 g/mm	

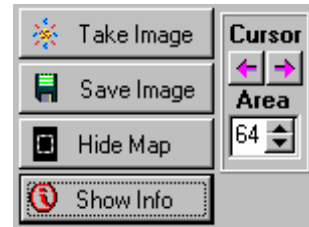
Spectral Mode

If you are in spectral mode, the primary window will display the current spectrum and the lower display window will show the last image taken, with or without outlines of the areas defined on the chip.



The options available for the lower window when in spectral mode include:

- **Take Image** takes a new image that will be displayed in this window.
- **Save Image** saves the current image in the window to a BMP, Tiff or SPC file.
- **Hide/Show Map** will trace the current map on the image in the window.
- The **Show/Hide Info** button toggles between the information screen shown above and the last image taken.
- The **Area** and **Cursor arrows** show the intensity and wavelength values for the specified area in the spectrum display window.

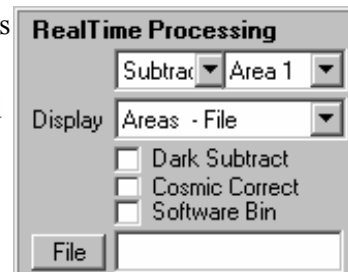


Real Time Processing panel

The Real Time Processing panel is used to do spectral manipulations on the data as it is being acquired and displayed.

Operation: The dropdown list choices for the top left field are used when doing mathematics between specified areas on the chip. The choices are NoOp (no operation), Divide, and Subtract. The dropdown list for the righthand field lists all of the current areas. The area specified in this field will be the area that will either be subtracted from or be divided into all of the current areas on the chip. If there are 4 areas and **Divide by** is specified in the left field and **Area 2** is specified in the right field, then the spectral display will show the results of **A1/A2**, **A3/A2**, and **A4/A2**.

Display: Incoming data can also be manipulated and displayed by using previously stored data. The Display field's dropdown list contains all of the display choices available based on the system configuration. Typical choices are: Areas, Areas/File, Areas - File, Absorbance(Areas),



Reflectance(Areas), and Transmission(Areas). Data from a stored file can be divided into or subtracted from all areas in the incoming data. Real-time calculations of absorbance, transmission, and reflection can be performed and displayed when a stored file is used for the reference.

File: This field allows you to retrieve data from a previously stored file. This data will then be used in the selected **Display** operation.

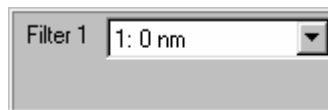
Dark Subtract: This check box is used to subtract the contribution of temperature dependent signal from a spectrum. Since the dark component is constant for any temperature and integration time, a spectrum of only the dark component can be subtracted from each spectrum. This is accomplished by taking a spectrum with the shutter closed and subtracting it from each subsequent readout. This has the effect in survey mode of doubling the time for each acquisition.

Cosmic Correct: This check box is used mainly with long exposures to eliminate any spurious peaks that may appear due to cosmic rays. When this is selected, two spectra with the specified integration time will be taken. If there are any peaks in one that are not in the other then these peaks will be removed from the displayed spectrum.

Software Bin: The Software Bin check box causes each row in the specified area to be individually read out and co-added in memory. This is fundamentally different from the normal readout where all the rows in the area are shifted into the chip's readout register and read out at once. The advantage of using software binning is that it allows for higher dynamic range in that individual rows may not over saturate the readout register, while the simultaneous readout of all of the rows might. The disadvantage to using this option is that the readout process takes significantly more time than in the normal hardware binning.

Filter panel

This panel appears if there is one or more filter wheel in your current experimental setup. The field shows the current filter position, or cut off wavelength, if specified, or name specified in the filter wheel set up tab in the hardware configuration screen. Click on the arrow to display all possible choices and highlight the desired filter.



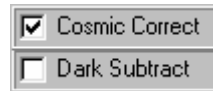
Trigger Status panel

The trigger status information panel shows whether user-defined triggers are active (green) or inactive (gray). To change the state of the triggers or to redefine the trigger ports or triggering criteria return to the Hardware Config screen and select the appropriate Triggers tab.



Cosmic Correction and Dark Subtraction panels

The Cosmic Correction and Dark Subtraction panels appear in imaging mode. These functions appear in the real time processing panel in spectral mode. Refer to the descriptions above.



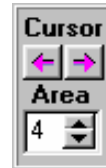
3d/Log panel

The 3d check box will cause successive spectral readouts to be displayed in a 3-dimensional plot in the primary window. The log check box will cause the intensity range to be displayed on a logarithmic scale.

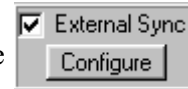


Cursor/Area Selection panel

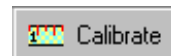
The Cursor/Area Selection panel only appears in spectral mode. It is used to select from which area the cursor will display the intensity value in the cursor coordinates panel at the top of the screen. Do not enter a value larger than the number of areas collected on the current acquisition.

**External Sync panel**

The external sync check box is used to synchronize the start of a readout with an external device. There must be the correct connection between the external device and the CCD detector or its ST-133 controller (if the detector has an external controller). The Configure button opens the Hardware Config screen and the CCD Triggers tab where you can set up External Sync triggering, logic output, shutter mode, and/or chip cleaning.

**Calibration**

Wavelength (or wavenumber) calibration of a spectrograph takes many parameters into consideration; the focal length of the instrument, the angle of the focal plane relative to the plane of the CCD chip, the difference in the spectrograph's dispersion at different center wavelengths, and the grating groove density. With SpectraSense once the calibration is computed for any central wavelength, it will remain valid for all central wavelengths. However, a separate calibration function must be created for each grating in the spectrograph. Once the gratings are calibrated, the software knows which function to apply for each grating when they are interchanged.

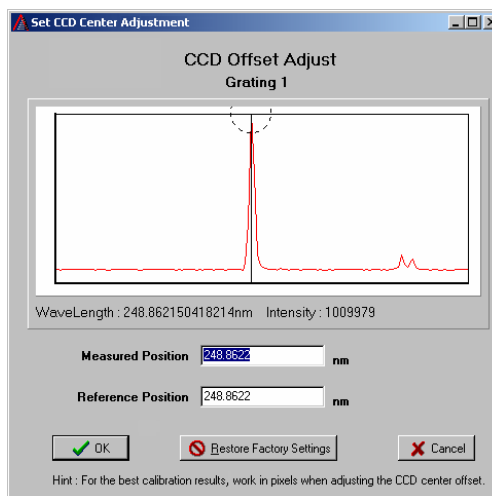
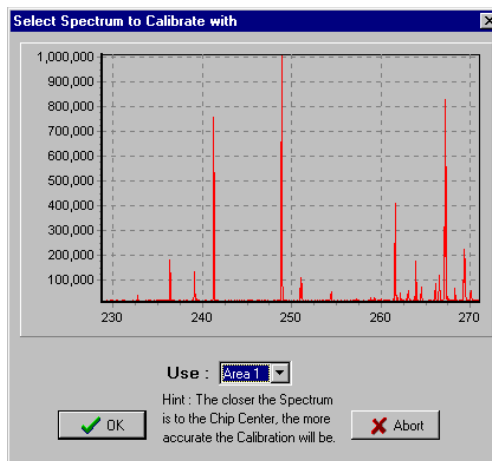
**To Perform a Calibration**

1. Set up a known wavelength standard to measure. Typically mercury pen lamps or fluorescent lights are used for calibration as they have very sharp, well known peaks. Usually the 435.8 nm peak or the 546.1 nm peak is used in the calibration procedure.
2. In the Hardware Config screen, select the grating that you want to calibrate.
3. Return to the Survey Mode screen and click the image/spectrum button so that you will be collecting spectra.
4. Illuminate the entrance slit of the spectrograph with light from your calibration source and take a spectrum with good signal to noise with the spectrograph centered at your reference peak.
5. Click on the **Calibrate** button. This opens the introductory dialog box for the calibration procedure. Note that you will repeat the procedure three times to accurately calibrate your system. Start out by selecting Step 1 and then click on the **OK** button. If at any time you feel that you made an error you can always click on the **Restore Factory Defaults** button.



6. You will come to a screen that shows the spectrum or spectra from multiple areas that you have just acquired. Select one area to work with **Use** dropdown list. Click on the **OK** button to continue.
7. On the third screen, expand the spectrum around your reference peak so that you can easily place the cursor exactly on the center of the peak.
8. Select the reference peak. If the reference peak is the highest peak in the spectrum, the software will automatically move the cursor to it. If the reference peak is not the highest peak, move the cursor to it and enter the wavelength value displayed below it into the **Measured Position** field. In general, there is no need to enter the value to more than two decimal places.
9. Next, enter the know wavelength of the peak into the **Reference Position** field and click on **OK**. This returns you to the Survey Mode screen.
10. Now, move the spectrograph so that the reference peak is located very near the left-hand edge of the spectrum.
11. Click again on the **Calibrate** button. Select Step 2 and repeat the procedures described above.
12. Next move the spectrograph such that the peak is now located near the right edge of the spectrum, select Step 3, and repeat the above procedures.
13. After you have completed this three-step process your spectrograph is calibrated for the selected grating. Change the grating and repeat the process again. If you have a third grating in your system, perform the calibration procedure for it also.

Once your system is calibrated, it should remain so unless you move the CCD detector head or if there is a noticeable change in ambient temperature. It is a good laboratory practice to verify the calibration from time to time by running a spectrum of your calibration source and noting the displayed value for the reference peak.



Menu Options

File

Open	Open saved chip area map
Save	Image
	Saves the current image to a file.
	Spectra
	Saves the current spectrum to a file.
	Map
	Saves the current area map to a file.
Print	Frame
	Sends a bitmap image of the screen to the printer.
	Image
	Prints a picture of the image displayed.
	Spectra
	Prints a plot of the spectrum or spectra displayed.
Print Setup	Calls Windows printer control screen.
Exit	Closes SpectraSense.

Edit

Copy Chart to Clipboard	Copies the image displayed or the spectrum or spectra displayed to the Windows clipboard.
Export Excel	Places data into a new Excel work sheet.
Export to Grams	Puts the data in the active working window in GRAMS software.
Export to Grams 3D	Exports multi-file data into GRAMS/3D [®] program. Program must be running concurrently with SpectraSense.
Overlay Spectra	Recalls a stored spectrum to be displayed concurrently with incoming spectra.

Select Tab

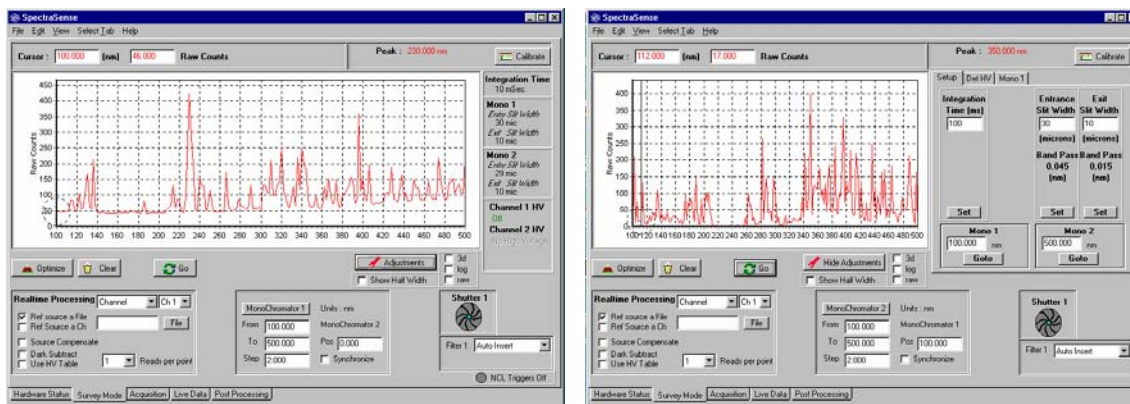
Hardware Config	Calls up the Hardware Config screen.
Slit Band Pass	Calls up the Monochromator Slit BandPass screen.
Survey Mode	Calls up the Survey mode.
Acquisition	Calls up the Acquisition screen.
Live Data	Calls up the Live Data screen.
Post Processing	Calls up the Post Processing screen.
Mono Control Software	Calls up the Acton Monochromator Control Software screen.
Terminal	Calls up terminal mode for communications debugging. Used for diagnostics in communications between the hardware and computer.

Help

Contents	Hyperlinked index of the SpectraSense manual.
Overview of SpectraSense	Complete SpectraSense Manual on line.
Using the CCD Survey Page	Context-sensitive help for this chapter.
SpectraSense Version	The version number of the installed software.

This page intentionally left blank.

Chapter 4: Scanning Survey Screen



Scanning Survey Mode

The Scanning Survey Mode Screen is used to optimize the acquisition parameters when acquiring data using a single channel detector such as a silicon cell, PMT, or IR detector. This screen will appear when the Survey Mode tab is selected if there is a single channel detector in the light path specified in the Hardware Config screen.

The spectral display screen shows the incoming data as it is being acquired. As previously described, the display can be expanded or contracted by dragging the mouse over the desired section of the spectrum.

Certain parameters must be defined before an acquisition can occur. There must be starting point and ending point of the acquisition, there must be step increment to tell the spectrometer how far to move between data points, there must be an integration time to define how long the signal will be collected at each data point, and there must be a definition of which detector or detectors are going to be used for acquiring the spectrum and what real time manipulations of the data are required. It is in this screen that these parameters are adjusted and optimized to produce spectra with the desired signal to noise ratio and acquisition time.

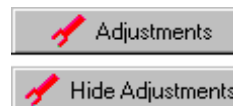
Scan Range Selection panel

The scan range selection panel is used to enter the start and end limits for the scan. The step is how many spectral units (nm in this case) the grating will turn before data is again taken. If there are two monochromators in your system, you can select which one will be scanned by clicking on the Monochromator button which will toggle between Monochromator 1 and 2. You can specify a specific position for the other monochromator if desired, and can have it scan an equal number of steps as the primary monochromator starting at the wavelength entered in the position box by checking the **Synchronize** box.

MonoChromator 1		Units : nm
From	100.000	MonoChromator 2
To	500.000	Pos 0.000
Step	2.000	☐ Synchronize

Adjustments button

The two-state adjustments button displays or hides the tabs that allow you to access the parameters that will affect the signal to noise quality of your spectra. Click on this button to access or hide these tabs.



Setup tab

The **Setup tab** is where the integration time for each data point is specified. After entering a value (in milliseconds) click on the set button to validate your entry. The longer the integration time, the better the signal to noise ratio, and the longer the time to collect a spectrum. If your monochromator has motorized entrance and exit slits, you can adjust the widths by entering values in the slit width fields and pressing on the **Set** buttons to validate. The values entered into these fields are stored with the spectrum.

Note: The slit information for motorized slits is stored with the acquisition parameters. The slit widths for manually adjusted slits are not stored.

You can quickly set the monochromators to a specific wavelength by entering the desired value and clicking on the **Goto** button. The position field will change color while the monochromator is moving.

 The screenshot shows the 'Setup' tab selected in a window with tabs 'Setup', 'Det HV', and 'Mono 1'. The 'Integration Time (ms)' field contains '100' with a 'Set' button below it. To the right, 'Entrance Slit Width' is '30 (microns)' and 'Exit Slit Width' is '10 (microns)', each with a 'Set' button. Below these, 'Band Pass' is '0.045 (nm)' and '0.015 (nm)', each with a 'Set' button. At the bottom, 'Mono 1' shows '100.000 nm' with a 'Goto' button, and 'Mono 2' shows '500.000 nm' with a 'Goto' button.

Det HV tab

If your system includes a photomultiplier tube, you can adjust the high voltage setting. Within a certain high voltage range, specified by the manufacturer of the specific tube, increasing the high voltage will increase the signal. You change the high voltage by entering a new value in the **Volts** field and verifying it by clicking on the **Set** button. If you want to change the value you have entered, click on the **Cancel** button and the previous value will return. You can manually turn the high voltage on or off by clicking on the **On/Off** button on this panel. Clicking on the Channel **Read** buttons will display the intensity at the current wavelength using the current integration time. You will note that Channel 2 does not have a high voltage associated with it. It may be a silicon cell or other solid state detector. None-the-less clicking on the **Read** button will display the intensity at the current position and integration time.

 The screenshot shows the 'Det HV' tab selected in a window with tabs 'Setup', 'Det HV', and 'Mono 2'. Under 'Channel 1 High Voltage', the 'Volts' field contains '329', with 'Set' and 'Cancel' buttons below it. A green indicator light is on, with an 'Off' button next to it. At the bottom, 'Channel 1' and 'Channel 2' each have a 'Read' button.

Mono tab

The number of Mono tabs depends on the number of monochromators in your experimental setup. It is from this tab that you can change the slit widths, if motorized slits are included in the spectrometer or you can change the grating if the grating turret has multiple gratings. You will note that on this tab the default setting units are in spectral band pass (i.e., you specify the desired spectral band pass and the slits will open to the appropriate widths). As spectral band pass varies with wavelength, the slits will automatically adjust to maintain the specified values. You specify if you want to work in microns or band pass in the Hardware Config screen. It is not recommended that you select the band pass option if your slits are manually adjustable.

The screenshot shows the 'Mono 1' tab with the following settings:

- Entrance Slit Width:** 25 (microns)
- Exit Manual:** 0.079 (nm)
- Mono 1:** 539.995 nm
- Grating:** 3 600 g/mm, Blaze =

You can also reposition the monochromator to a new central wavelength by entering the value in the **Mono** panel and clicking on the **Goto** button.

Real Time Processing panel

The Real Time Processing panel is used to define which detectors are to be read out for display and how the incoming data is to be processed before it is displayed.

Display options: The dropdown list for the top left field contains all of the display options available, which will vary depending on your system configuration. Typical choices are: Channel, Ch/Ref, Ch-Ref, Absorbance, Reflectance, Transmittance, and All Channels. The channel number is chosen from the dropdown list of the righthand field. There must be a reference for the Ch/Ref, Ch - Ref, Absorbance, Transmission, and Reflectance measurements. This reference can be either a stored file or the output from another detector.

The screenshot shows the 'Realtime Processing' panel with the following settings:

- Ch - Ref:** Ch - Ref
- Ch 1:** Ch 1
- Ref source a File:** ☒ (File button visible)
- Ref Source a Ch:** ☐
- Source Compensate:** ☐
- Dark Subtract:** ☒ (Reads per point: 2)
- Use HV Table:** ☐

Ref source a File: When checked, a stored data file will be used as the reference for the channel/reference, channel - reference, absorbance, transmission, and reflectance measurements. If this box is checked, the **File** button will appear so you can browse for and select the file to be used as the reference.

Ref Source a Ch: When checked, a detector will be used as the reference. If this box is checked, a field will appear in which you specify the channel to be used as the reference.

Source Compensate: When checked and a source compensation accessory is included in the system, each data point will be divided by the signal from this accessory.

Dark Subtract: When checked, dark subtraction is turned on. A new dark signal will be subtracted from each reading. (When you check this box, the **Do Not Take New Dark Reading** check box is displayed to the right of the **Go** button.) All detectors produce a signal output that is temperature dependent. This signal can be subtracted from all readings when the detector is making a measurement. The software will close any automated shutter and take an initial reading or, if the last position on the filter wheel has been specified for dark subtraction, the wheel will

move to this position, take a reading, and then return to the appropriate position for the scan. If your system has neither a shutter nor filter wheel, you will be asked to block the light from entering the monochromator by any means of your choice, the dark reading will be taken, and you will be asked to unblock the light before the actual scan is taken.

Do Not Take New Dark Reading: When checked, a new dark reading will not be taken each time data is acquired ----the most recently acquired dark reading will be used until you uncheck the box.

Use HV Table: When checked, the current HV Table is used to automatically adjust the high voltage on the photomultiplier tube as the monochromator is scanned. Before using this option, make sure that there is an HV table. An HV table can be created or loaded on the Survey Mode Det HV tab.

Reads per point: The **Reads per point** field allows you to specify one or more reads per data point. If the signal is extremely weak, at times it is reasonable to average several readouts at each data point. For example, if 3 is entered, the monochromator will sit at each position and take three readings at the specified integration time and then output the average value from the three readings. For most cases one readout per point is sufficient.

Go/Stop button

After all of the parameters that are necessary to perform an acquisition are specified, it is only a matter of clicking on the **Go** button. The button changes to a **Stop** button once the scan is started.



Optimize/Stop button

After you have acquired your initial spectrum, it is possible to optimize the peak signal by using the **Optimize** button. In optimize mode the monochromator will move to the highest peak and start taking intensity readouts at the specified integration time. The display will look like a strip chart recorder where you can see in real time how the signal intensity changes as you modify acquisition parameters such as integration time, slit width, and high voltage. Once you have made the adjustments to your liking, click on the **Stop** button to go back to scanning mode. You can also set the monochromator to a specific wavelength by entering the value in the Mono panel of the **Set up** tab.



Clear button

The **Clear** button will clear the spectral display window. If the window is not cleared at the end of a scan, the subsequent scan will overlay on the previous one.



Cursor Information panel

The cursor information panel displays the current position and intensity information for all active detectors. The readout values (Raw Counts, Counts per Second, Amps, Volts, Counts Sec for Photon Counters) are selected on the Detection tab.



Display Format options

The display format options include 3D display of multiple spectra, the intensity displayed on a log scale, and the raw data input rather than any preprocessed data. The raw option is useful helping to determine the cause for spectra that do not appear as expected. It will show the raw data from all detectors on the same intensity scale.



Show Half Width check box

When checked, the wavelength position and spectral resolution of the highest peak on the spectral display window will be shown in the information panel at the top of the screen. To determine the spectral resolution of a smaller peak, expand the display such that that peak is the highest displayed.

☐ Show Half Width

Shutter and Filter panels

If your system has an automated shutter, its current status (i.e., open or closed) is shown in the **Shutter** panel. Click on the shutter icon to change the state of the shutter. If an automated filter wheel is present, the **Filter** panel will appear. The choice field shows the current filter in place or Auto Insert if this option has been defined on the Hardware Config screen. You can insert any filter by clicking on the choice arrow and highlighting the appropriate filter.

**Trigger Status panel**

The Trigger Status panel shows whether user-defined trigger are active (green) or inactive (gray). To change the state of the triggers or to redefine the trigger ports or triggering criteria return to the Hardware Config screen and select the appropriate Triggers tab.

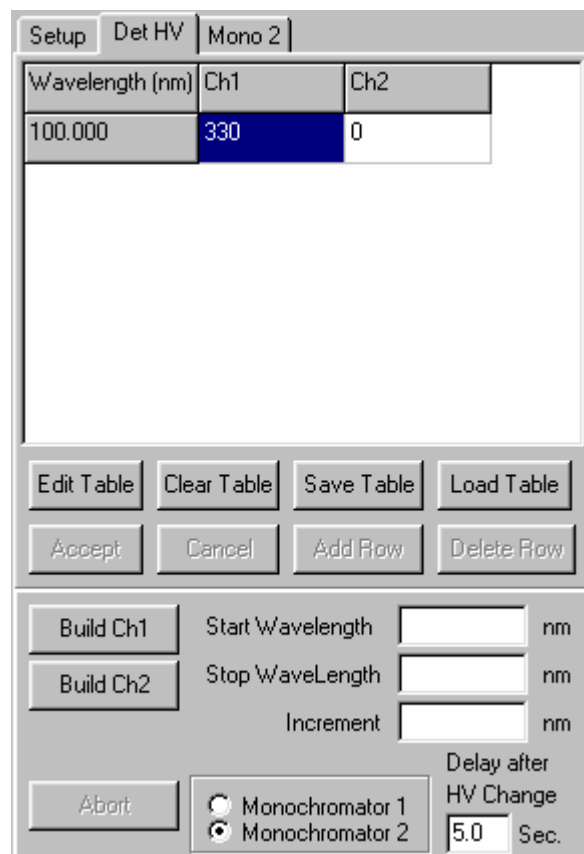
**High Voltage table**

When the **Use HV table** box is checked, the current HV table will be used to change the high voltage on the photomultiplier tube as the monochromator scans. This is to ensure that spectral response of the system remains within a reasonable working range. Initially, the table is empty and you will have to create one either by manually adding rows and values or using the autogenerate functions. After you have created an appropriate table, you can save it, use it for the next acquisition, or reload it at a later time.

To Autogenerate a High Voltage Table:

In order to create the table, a broad band light source in the region of interest must be placed at the entrance of the monochromator.

Note: For the SR-510 and compatible lock-ins, the HV table is used to provide the voltages (mV) for the AUX5 and AUX6 D to A output channels. The autogenerate table function is not available. Instead you must manually build the table using the Edit Table, Add Row, and Delete Row features.



1. Place a broad band light source at the entrance of the monochromator.
2. Check the **Use HV Table** box.

3. Click on the **Adjustments** button and select the **Det HV** tab.
4. If you have more than one monochromator, specify the monochromator to which the specific detector is attached in order for the program to display the appropriate input channels.
5. Specify the range over which you want the correction to be made by entering a Start and Stop wavelength. Enter an increment to specify how finely you want to make the adjustments. Enter a delay after HV change in the several seconds range to allow the photomultiplier tube to stabilize to the new voltage. These values and the channel HV (entered on the **Det HV** tab when the **Use HV Table** check box is unchecked or on the **Hardware Setup|Detection** tab) will be used to populate the column for the channel you select.
6. Click on the appropriate Build Ch# button to start the procedure.

Note: If there are already values in the table, you will be asked if you want to clear the existing contents of the table. If you click on Yes, all cells in the table will be cleared and the new column will be generated for the channel you selected. If you click on No, existing values in the column for the channel you selected will be overwritten by the new values and values in other columns will be retained.

7. After you respond to the dialog box, the table will automatically be generated by the software and the wavelength and HV values will be entered into the appropriate column.
8. Repeat Steps 5 and 6 for any other channels.
9. You can now edit the HV values in the table, clear the table, save the table to a file, or recall a previously stored table.

Menu Options

File

Save	Saves the current spectrum to a file.
Print Frame	Sends a bitmap image of the screen to the printer.
Spectra	Uses the TEE Chart print function to print a plot of the spectrum or spectra displayed.
Data	Prints out a text version of the data file, including parameters and data coordinates.
Print Setup	Calls Windows printer control screen.
Exit	Closes SpectraSense

Edit

Copy Chart to Clipboard	Copies the image displayed or the spectrum or spectra displayed to the Windows clipboard.
Export to Excel	Puts the data into an Excel worksheet.
Export to Grams	Puts the data in the active working window in GRAMS software.

View

Autoscale	Automatically scales the axes of the plot so that the full spectrum is shown.
Full Width	Show the spectrum with the current vertical expansion but full wavelength coverage.
Full Height	Shows the spectrum with full vertical expansion but with current horizontal expansion.
Zoom In	Expands the spectrum to show half the current values as full scale on the display.
Zoom Out	Shrinks the current expansion to twice the current level.
Overlay Spectra	This allows you to recall stored spectra and display them on the same axis as the most recently acquired spectrum in the display window.

Select Tab

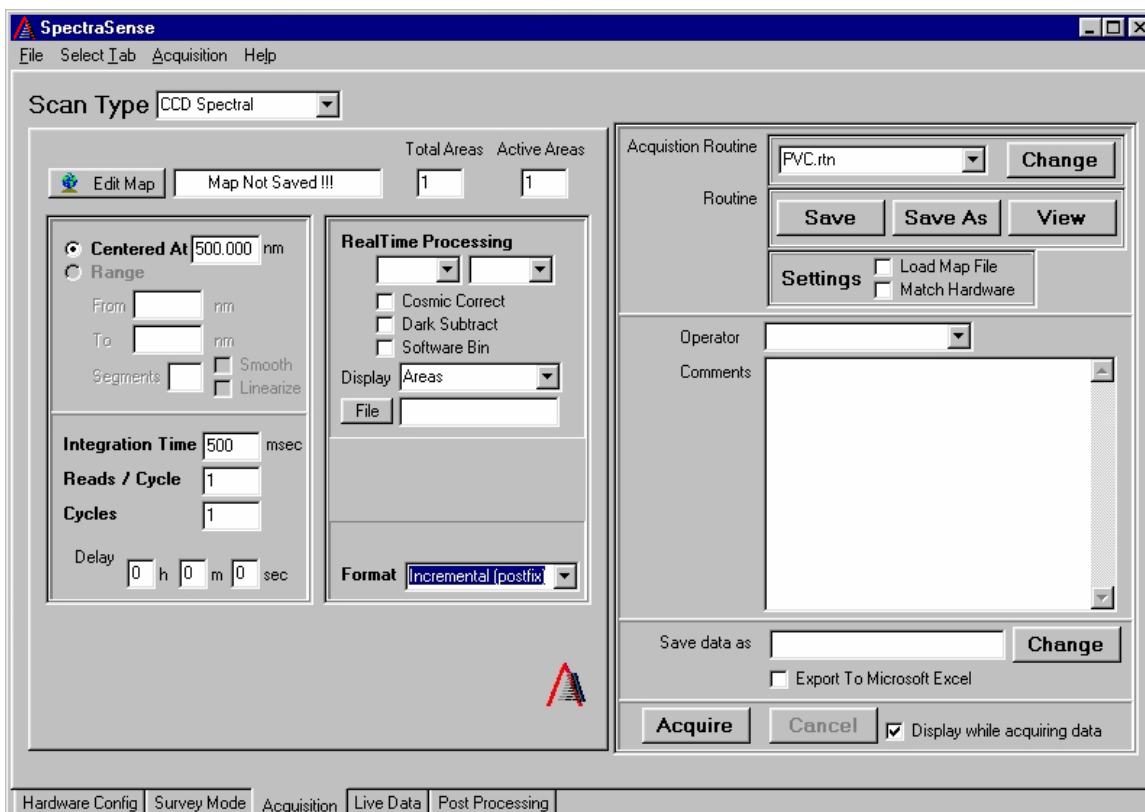
Hardware Config	Calls up the Hardware Config screen.
Slit Band Pass	Calls up the Monochromator Slit BandPass screen.
Survey Mode	Calls up the Survey mode.
Acquisition	Calls up the Acquisition screen.
Live Data	Calls up the Live Data screen.
Post Processing	Calls up the Post Processing screen.
Mono Control Software	Calls up the Acton Monochromator Control Software screen.
Terminal	Calls up terminal mode for communications debugging. Used for diagnostics in communications between the hardware and computer.

Help

Contents	Hyperlinked index of the SpectraSense manual.
Overview of SpectraSense	Complete SpectraSense Manual on line.
Using the Survey Page	Context-sensitive help for this chapter.
SpectraSense Version	The version number of the installed software.

This page intentionally left blank.

Chapter 5: CCD-Based Acquisition



CCD Spectral Acquisition

The Acquisition screen is used to collect spectra or time-based intensity at a specific wavelength measurement. The acquisition parameters that were optimized in the Survey Screen are carried over to the Acquisition screen to ensure that the data are collected with the desired signal to noise ratio and resolution. It is from these screens that routines can be created that will reproduce acquisition conditions that can be recalled at a later date. The acquisition mode associated with CCDs is **CCD Spectral Acquisition** and it can be accessed from the Scan Type field at the top of the screen. If your system also includes single channel detectors, the acquisition types associated with them will also be accessible from the Scan Type field's dropdown list. If you inadvertently choose an acquisition mode not associated with the detector in the current light path, the parameter input fields will not appear. If the acquisition mode you want to use will not come up, go to the Hardware Config screen and change the light path so the desired detector will be activated.

The Acquisition screen is set up so that acquisition parameters are entered on the lefthand side of the screen and file-related parameters are entered on the righthand side.

In the spectral acquisition mode, individual spectra may be acquired, multiple spectra collected with a specified delay between readouts can be acquired, multiple spectra “glued” together to produce a data file that covers an extended spectral range can be acquired, and excitation/emission profiles can be acquired. The left side of the acquisition screen is used to define how the spectrum or spectra will be acquired.

Acquisition Parameters

Many of the acquisition parameters are transferred from the Survey Mode screen but they can be changed here.

Area Map Information

This panel shows at a glance the total number of areas currently defined on the chip. It is possible to change

the size and number of areas after clicking on the **Edit Map** button. The area map creation panel will appear allowing you to either load previously stored area maps, modify the current area map, or change which areas in the current map are read out or ignored. Refer to the *"CCD Survey Screen"* chapter for more information on this subject.

CCD Spectral Position panel

In all cases, the spectrometer must be positioned to capture the spectral region(s) of interest. This information is entered into the position panel.

Centered At option: Select this option if the entire spectral region of interest can be dispersed on the CCD chip. Enter into the field the wavelength that will be in the center of the range.

Range option: Select this option if the spectral range to be covered is wider than can be captured by the CCD all at once.

When the range choice is checked you must enter a starting and ending wavelength to be covered. The next thing you need to do is decide how this range is to be divided into steps (i.e., the number of times the spectrometer will move and acquire incremental spectra that will then be "glued" together to create the dataset for the entire range).

Overlap/Steps: When you select **Overlap**, you then enter the percentage of overlap and SpectraSense will calculate the number of steps. When you select **Steps**, you enter the number of steps. You can estimate the number of steps by noting the wavelength coverage on a single readout and determining how many of these segments will be needed to cover the full range. It is advisable to allow an extra segment if the coverage is close. For example; if in the Survey screen you noticed that the spectrum covered the range from 200 to 400 nm, you can make the rough assumption that 200 nm will be covered in every segment. To cover the range from 300 nm to 800 nm would require 3 segments and would have sufficient overlap between segments to ensure continuity.

Steps button: Click on this button to see a list of the steps and the range that each step covers. The list is based on your selection and entries for Overlap/Steps.

Smooth check box: When checked, the intensity differences in the overlap regions will automatically be averaged to produce a smooth continuous spectrum.

Thin check box: When checked, smoothing is applied to the center 1/8th of the overlap regions instead of to the full overlap regions.

Linearize check box: When checked, the data is extrapolated so that the spacing between data points is in equal wavelength units as opposed to pixel spacing.

Readout panel

The Read Out panel is used to define the integration time for the spectral acquisition and how many spectra are to be collected.

Integration Time: Initially, this is the value used in the Survey screen, as under normal operation the integration time is optimized in the Survey screen. This can be changed by enter a new value into this field.

Reads/Cycle: The Reads/Cycle field determines how many readouts will be averaged together for each spectrum. In general, one readout is sufficient. In cases where the integration time is long, it may be useful to coadd reads to reduce the chances of cosmic-induced peaks.

Cycles: The Cycles field determines how many spectra are to be acquired in the experiment.

Single Start (No Live Display): Single Start means one start for all cycles versus a start for each cycle. The result is faster data acquisition with no live data display.

Average Cycles: This check box is active when there are multiple cycles. When checked, the spectra from the cycles will be acquired and coadded. The result will be divided by the number of cycles and the averaged spectrum will be stored in a single data file.

Delay: The Delay (in hours, minutes, and seconds) determines the time between each acquisition. The delay includes the integration time; if the integration time is 500 ms and the delay is 600 ms, the camera will integrate the signal for 500 ms and wait an additional 100 ms before starting the next acquisition.

Integration Time 100 msec
 Reads / Cycle 1
 Cycles 10
☐ Single Start (No Live Display)
☐ Average Cycles
 Delay 0 h 15 m 59 sec

Real Time Processing panel

The incoming data from the CCD can be pretreated before being displayed and saved by selecting options in the Real Time Processing panel.

Area Divide by/Subtract: If more than one area of the chip is being read out, it is possible to either divide the intensities of all other areas by one, or to subtract the spectrum from one area from all of the others. These options are initiated by selecting No op, Divide by, or Subtract from the top left field's dropdown list and selecting one of the areas from the top right field's dropdown list. If there is only one area being used, these options will not be accepted and an error message will appear.

RealTime Processing
 Divide by Area 1
☐ Cosmic Correct
☐ Dark Subtract
☒ New Dark
☐ Software Bin
 Display Areas - File
 File

Cosmic Correct: When checked, the incoming data can be corrected for cosmic interference. Two acquisitions with the defined integration time will be taken and the two spectra will be compared. Any peak not found in both spectra will be removed. This option should only be used for long integration times, usually above 60 seconds.

Dark Subtract: When checked, a spectrum will be taken with the shutter closed. This spectrum will then be subtracted from all subsequent readouts. If any CCD parameters (such as integration time, center wavelength, or range) have changed since the last background was taken, a new background is taken.

New Dark: When checked, a new background is taken at the start of the acquisition. If unchecked, the last stored background will be used for background subtraction.

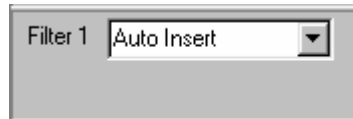
Software Bin: When checked, each row of pixels in the active area(s) will be read out individually and have the intensities from all of the rows coadded in memory. This will result in a noticeable delay in the acquisition but may eliminate saturation of strong peaks.

Display: Incoming data can also be manipulated and displayed by using previously stored data. The Display field's dropdown list contains all of the display choices available based on the system configuration. Typical choices are: Areas, Areas/File, Areas - File, Absorbance(Areas), Reflectance(Areas), and Transmission(Areas). Data from a stored file can be divided into or subtracted from all areas in the incoming data. Real-time calculations of absorbance, transmission, and reflection can be performed and displayed when a stored file is used for the reference.

File: This field allows you to retrieve data from a previously stored file. This data will then be used in the selected **Display** operation.

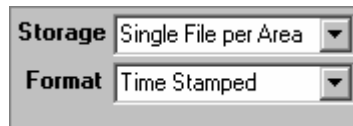
Filter Wheel panel

If one or more filter wheels are present in the system optical path, the filter panel will appear. The current state of the filter wheel is displayed in the choice field. Clicking on the arrow will present all of the installed filters and allow you to select which one to use in this acquisition.



Format/Storage panel

Storage field: Active when you are saving the data to an .SPC file. The Storage field allows you to choose either **Single File per Area** or **Single File per Cycle**.



Single File per Area will save the data in a format compatible with GRAMS/32. If you have two areas and one cycle, two files will be created. If you have two areas and three cycles, six files will be created.

Single File per Cycle will save the data in a format compatible with GRAMS/32 and GRAMS/3D. If you have two areas and three cycles, three files will be created and each file will contain two spectra (one per area).

Format field: Allows you to specify whether files from the same acquisition program are specified with a numerical prefix, a numerical suffix, or a Time Stamp suffix.

Examples:

Numerical Prefix: 001myfile.arc_data - ###myfile.arc_data

Numerical Suffix: myfile001.arc_data - myfile###.arc_data

Time Stamp Suffix: myfile_Area01_5-9-2006_2h41m22s_pm.arc_data

Excitation Profile panel

The Excitation Profile panel will appear if your system includes two spectrometers and you are using one in conjunction with a light source for fluorescence excitation. The fields and check boxes within this panel change depending on the **Excitation Profile** check box status and whether the spectrum (or spectra) are centered or include a range of steps.

Centered Spectra: If the spectrum/spectra are centered on a particular wavelength, the panel will either contain one field or three fields depending on the status of the **Excitation**

Profile check box. If the box is unchecked, only the wavelength position of the excitation monochromator will appear in a field named **Center**. Entering a value in this field will cause the monochromator to go to that wavelength before all acquisitions are made. If the box is checked, three fields will appear. An excitation profile takes a spectrum or series of spectra as defined above, for each excitation wavelength defined in this panel. The results can be viewed as a 3D surface plot using programs such as GRAMS/3D. The excitation range is defined by the **From** and **To** limits entered into these fields. The **Steps** field defines how many wavelength units (nm in this example) the excitation monochromator will move before the next spectrum or series of spectra will be taken. In this example (400-560 nm, 10 nm steps), 16 spectra will be taken in the profile.

Stepped Range: If more than one step is required to cover the spectral range, this panel will contain at least one more check box and possibly a field. **Sync Move** allows you to synchronize the two spectrometers for the step and glue operation. If you want to include a specific offset between the two spectrometers and you have checked **Sync Move**, click in **Use Offset** and enter the value in the **Offset** field.

The image shows three screenshots of the 'Excitation Profile' panel. The top panel shows the 'Excitation Profile' checkbox unchecked, with a 'Center' field set to 400.000 nm. The middle panel shows the 'Excitation Profile' checkbox checked, with 'From' set to 400.000 nm, 'To' set to 560.000 nm, and 'Steps' set to 10 nm. The bottom panel shows the 'Excitation Profile' checkbox unchecked, 'Sync Move' checked, and 'Use Offset' checked, with an 'Offset' field set to 150.000 nm.

*Note: When the **Sync Move** check box is present, checking the **Excitation Profile** box, will change the spectral location from **Range** to **Centered At**.*

File Parameters

The right hand side of the screen is used to save all of the parameters entered in the Acquisition panel into a file (*.rtn) that can be recalled at any time to collect spectra under identical instrumental conditions.

Routines can also be recalled in this panel to override the current hardware and operating parameters with those defined in the stored routine file. Although they are not present in this screen, the grating choice and slit width settings used in the original acquisition can also be recalled and set. Data can be stored to a file in either .data, .SPC, or .txt formats or directly exported into an Excel spreadsheet, GRAMS/32, or GRAMS/3D depending on the file type and acquisition setup.

Routine panel

The Acquisition Routine panel is used to recall or create new acquisition routines. The field will display "Routine Note Saved!!!" if the current conditions have not been recalled from a stored routine file. The most recently used stored routines are shown as choices in the panel.

Change: If the desired routine is not displayed here, clicking on the **Change** button will allow you to browse through your directories and recall the desired routine.

Save: Stores the currently displayed acquisition parameters and hardware status such as grating used and slit widths in a new routine file.

Save As: Click on this button if you have recalled a routine, modified it and want to preserve the original routine and save the modified routine.

View: Opens a window that shows all of the instrumental parameter settings used when the routine was stored.

Settings panel

Load Map File: You may want to reproduce the acquisition parameters defined in a routine but not necessarily all of the instrumental parameters. If you do not care if the area map is identical to the one stored with the routine you can uncheck Load Map File. If the area map settings are critical, check the box to ensure that the same area map is loaded before the acquisition starts.

Match Hardware: Hardware parameters that can be matched or ignored include the grating choice and slit width settings. If the Match Hardware box is checked, any differences between the current setting and stored settings will be noted in a pop up dialog box.

Operator/Comments panel

Operator field: Use this field to store the name of the person who acquired the spectra in the file information header. The dropdown list contains the names entered as default users via the **Acquisition Preferences** menu item.

Comments field: Use this field to store comments or other experimental conditions in the data file.

Save panel

Save data as field: Enter the file name you want to use to store your spectra in this field.

Click on the **Change** button to bring up the

Save Data As... dialog box where you can also select a different file format: arc_data, SPC, or txt.

Export to Microsoft Excel: When checked, the data will also be sent to Microsoft® Excel®. The number of Excel spreadsheet files generated is based on the number of active areas and the number of cycles. If you have four areas and five cycles, twenty Excel files will be created.

Export to Grams/32: Available when the SPC format is selected. When checked, data will be exported to GRAMS/32. Refer to "**Chapter 10: Importing Data into G**" for more information.

Export to Grams 3D: Available when the SPC format is selected and Single File per Cycle is the **Storage** mode. When checked, data will be exported to GRAMS/3D. Refer to "**Chapter 10: Importing Data into G**" for more information.

Display while acquiring data: When checked, the acquisition will be displayed in real time on the Live Data screen. If the acquisition time is very short and the delay between reads is also very short, the data may not be displayed in real time so as to not affect the timing of the experiment. Also, if the **Single Start (No Live Display)** box is checked, the data will not be displayed as it is being acquired.

Acquire: Click on this button to start the acquisition. If you have not specified a file name, a dialog box will appear asking if you want to continue or enter a name. You may stop an acquisition at any time by clicking on the **Cancel** button.

Menu Options

File

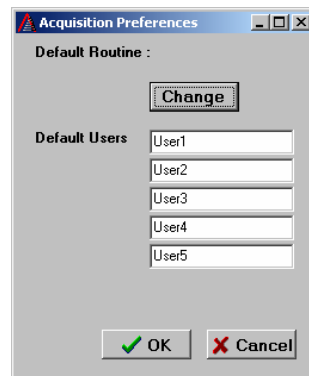
Exit Closes SpectraSense

Select Tab

Hardware Config	Calls up the Hardware Config screen.
Slit Band Pass	Calls up the Monochromator Slit BandPass screen.
Survey Mode	Calls up the Survey mode.
Acquisition	Calls up the Acquisition screen.
Live Data	Calls up the Live Data screen.
Post Processing	Calls up the Post Processing screen.
Mono Control Software	Calls up the Acton Monochromator Control Software screen.
Terminal	Calls up terminal mode for communications debugging. Used for diagnostics in communications between the hardware and computer.

Acquisition Preferences

Preferences... Opens the Acquisition Preferences dialog where you can specify a default routine to be loaded each time that SpectraSense is run. You can also enter the names of up to five default users. These names will be inserted into the Operator dropdown list so the appropriate name can be selected for storage with the data.



Help

Contents	Hyperlinked index of the SpectraSense manual.
Overview of SpectraSense	Complete SpectraSense Manual on line.
Using the Acquisition Page	Context-sensitive help for this chapter.
SpectraSense Version	The version number of the installed software.

Chapter 6: Scanning Spectral Acquisition

The screenshot shows the SpectraSense Acquisition window. At the top, the 'Scan Type' is set to 'Single Point Scan Mono1'. The 'Monochromator 1' section includes wavelength settings (From: 450.000 nm, To: 500.000 nm, Step: 2.000 nm) and channel settings (Channel 1: 500 nm, Channel 2: 350 nm, Channel 3: 0 nm). The 'Realtime Processing' section has checkboxes for 'Ch / Ref', 'Ref Source a File', 'Ref Source a Ch', 'Source Compensate', 'Dark Subtract', 'Do Not Take New Dark Reading', and 'Use HV Table'. The 'Excitation Profile' section is checked, showing 'From: 1.000 nm', 'To: 10.000 nm', and 'Steps: 1.000 nm'. The 'Storage' section is set to 'Single File per Scan' and 'Format' is 'Average Cycles'. The 'Acquisition Routine' section shows 'Routine Not Saved !!!' and buttons for 'Change', 'Save', 'Save As', and 'View'. The 'Operator' is 'Ferris Hanready' and the 'Comments' field contains 'Sample heated to 350 for 2 hours.'. The 'Save data as' field is 'Test.arc_data' and 'Export To Microsoft Excel' is unchecked. The 'Acquire' and 'Cancel' buttons are at the bottom. The status bar at the very bottom shows 'Hardware Config', 'Survey Mode', 'Acquisition', 'Live Data', and 'Post Processing'.

Single Point Scan Mono

When a Single Point Scan mode is selected, the Acquisition screen is used for collecting spectra from single channel detectors such as photomultiplier tubes, silicon cells, and most solid state infrared detectors. In the normal progression in setting up and acquiring good spectra, the Acquisition screen is usually employed after the parameters that affect spectral quality are optimized in the Survey Mode screen. For the most part, all of the previously adjusted parameters are automatically transferred over to this screen. It is from this screen that single spectral experiments as well as those requiring multiple spectral acquisitions can be defined and run.

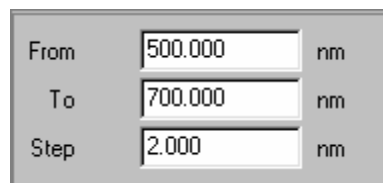
If there are two monochromators in your system there will be two scan choices, one for Mono 1 and one for Mono2. This allows you to set up two discretely different experiments.

The Acquisition screen is also used to create and recall experimental routines. A routine contains all of the acquisition parameters and hardware settings to exactly reproduce spectral acquisition conditions. Creating routines for often repeated experimental procedures ensures that all spectrometric conditions will be identical and that all of the information on the system parameters is safely stored within the data file.

Acquisition Parameters

Scanning Parameters panel

The scanning parameters specify the spectral range to be covered and the increment between data points. When you open this screen the last values used in the Survey mode are loaded into the fields. You have the option of changing any or all of the parameters by highlighting a field and entering new values. If you recall a routine, these values will be changed to those stored in the routine. As a rule of thumb, if you know the spectral resolution of the narrowest peak you will be measuring, set the Step to approximately 1/5 that value. For example, if the narrowest peak you expect to see in your spectrum is 10 nm wide at half maximum intensity, set the step to 2 nm. Fewer steps may cause loss of resolution and data. Very much smaller steps will, for the most part, only make the acquisition time longer and the data file larger.



From 500.000 nm
To 700.000 nm
Step 2.000 nm

Data Acquisition panel

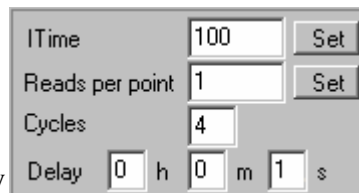
Note: If there is a Set button associated with a parameter, click on that button after you have made your choice.

ITime: The integration time is used to optimize your signal to noise ratio in the Survey screen is automatically transferred to the ITime field. You can change the integration time by highlighting the field and entering a new value.

Reads per point: The **Reads per point** field allows you to specify one or more reads per data point. If the signal is extremely weak, at times it is reasonable to average several readouts at each data point. For example, if 3 is entered, the monochromator will sit at each position and take three readings at the specified integration time and then output the average value from the three readings. For most applications it is advisable to set the integration time and slit widths wide enough so that there is good signal to noise on a single readout of the detector. Multiple readouts may be necessary under conditions where the signal is so weak that it is just distinguishable from the noise.

Cycles: This field is used to indicate how many times spectra are to be collected during the experiment. Each spectrum can be saved individually or all of the cycles can be signal averaged together to create one file.

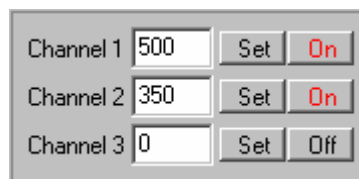
Delay: Enter the delay in hour, minutes, and seconds between cycles (spectra). The delay must be longer than the time to acquire one spectrum and slew back to the starting wavelength. The time to acquire a spectrum and return to the starting wavelength depends not only upon the integration time and number of increments, but also the time to move between data points and the time to slew back. These parameters change under different acquisition conditions and can not be default calculated. It is advisable to take a single scan and measure the time it takes to acquire. The time to slew back is normally 10 seconds or less. The combination of these represents the minimum allowable delay between cycles.



ITime 100 Set
Reads per point 1 Set
Cycles 4
Delay 0 h 0 m 1 s

High Voltage panel

The High Voltage panel will appear if there are photomultiplier tubes or an SR-510 in your system. The current value is that last used in the Survey mode. You may change the high voltage by highlighting the field and entering a new value. Click on the Set button to validate the change. If a stored routine is loaded, the



Channel 1 500 Set On
Channel 2 350 Set On
Channel 3 0 Set Off

high voltage will be changed to the value stored in the routine. You can turn the high voltage on or off by clicking on the On/Off button.

Real Time Processing panel

The Real Time Processing panel will normally have values in all of the fields and appropriate check boxes checked. These values are carried over from the Survey Screen or reloaded from a stored routine. You can however change any of the values or check box options. The Real time Processing panel is used to define which detectors are to be read out for display and how the incoming data is to be processed before it is displayed.

Display: The dropdown list for the top left field contains all of the display options available, which will vary depending on your system configuration. The typical choices are: Channel, Ch/Ref, Ch-Ref, Absorbance, Reflectance, Transmission, and All Channels. If you are dividing by or subtracting a reference or have selected Absorbance, Reflectance, or Transmission, you must also select the reference source.

Channel Number: The channel number is chosen from the field to the right of the Display field. With the exception of All Channels, you must select a channel if there are multiple channels.

Ref Source a File: When checked, a stored data file will be used as the reference for the absorbance, transmission, and reflectance output.

Ref Source a Ch: When checked, the specified detector channel will be used as the reference.

File/Channel field: The purpose of this field depends on your check box selection. If the reference source is a file, it allows you to browse for the file and select the one you want used. If a channel is the reference source, this field allows you to select the channel.

Source Compensate: When checked and a source compensation accessory is included in the system, each data point will be divided by the signal from this accessory.

Dark Subtract: When checked, dark subtraction is turned on. A new dark signal will be subtracted from each reading. All detectors produce a signal output that is temperature dependent. The software will close any automated shutter and take an initial reading or, if the last position on the filter wheel has been specified for dark subtraction, the wheel will move to this position and take a reading and then return to the appropriate position for the scan. If your system has neither a shutter nor a filter wheel, you will be asked to block the light from entering the monochromator by any means of your choice, the dark reading will be taken, and you will be asked to unblock the light before the actual scan is taken.

Do Not Take New Dark Reading: When checked, a new dark reading will not be taken each time data is acquired ----the most recently acquired dark reading will be used until you uncheck the box.

Use HV Table: When checked, the current HV Table is used to automatically adjust the high voltage on the photomultiplier tube as the monochromator is scanned. Before using this option, make sure that there is an HV table. An HV table can be created or loaded on the Survey Mode Det HV tab.

Excitation Profile panel

The Excitation Profile panel will appear if there are two monochromators in the system.

Excitation Profile: When checked, a series of fluorescence emission spectra will be taken at different excitation wavelengths. For each excitation wavelength there will be an emission spectrum taken with the parameters defined in the panels above. The data is usually displayed as a 3D surface plot.

From: Specify the shortest excitation wavelength desired.

To: Specify the longest excitation wavelength.

Steps: Specify the excitation wavelength step increment.

Run Backwards: Check this box if your sample is subject to photo degradation at short wavelengths.

Synchronized: When checked, the excitation monochromator will move one step increment in lockstep with the emission monochromator. You specify the starting wavelength in the From field and the step increment in the Steps field. The number of steps taken will be the same number of steps taken by the emission monochromator.

If neither the Excitation Profile nor Synchronized box is checked, the monochromator will move to the specified wavelength and stay there through all of the spectral acquisitions specified in the panels above.

Filter Wheel panel

The filter wheel panel appears if there is a filter in your system.

The filter number or auto insertion option will appear in the choice field. You may change the filter by clicking on the arrow and highlighting the desired filter.

Format/Storage panel

Storage field: Active when you are saving the data to an .SPC file. The Storage field allows you to choose either **Single File per Scan** or **Single File, All Scans**.

Single File per Scan: Saves the data from each cycle into a separate file. If you have three cycles, three files will be created.

Single File, All Scans: Saves the data from each cycle into the same file. If you have three cycles, the generated file will contain three spectra.

Format field: Allows you to specify whether files from the same acquisition program are specified with a numerical prefix, a numerical suffix, average cycles, or a time stamp suffix. If you enter a base filename of "myfile", select the ARC data format, and select Average Cycles, the average cycles option will produce a single data file named myfile.arc_data that contains the averaged data from all of the cycles. Using the same base filename, examples of the other format filenames are:

Numerical Prefix: 001myfile.arc_data - ###myfile.arc_data

Numerical Suffix: myfile001.arc_data - myfile###.arc_data

Time Stamp Suffix: myfile_001_5-9-2006_2h41m22s_pm.arc_data

File Parameters

The File Parameters panel is used to save all of the parameters entered in the Acquisition panel into a file (*.rtn) that can be recalled at any time to collect spectra under identical instrumental conditions. Routines can also be recalled in this panel to override the current hardware and operating parameters with those defined in the stored routine file. Although they are not present in this screen, the grating choice and slit width settings used in the original acquisition can also be recalled and set. Data can be stored to a file in either arc_data, SPC, or txt formats or directly exported into an Excel spreadsheet.

Routine panel

The Acquisition Routine panel is used to recall or create new acquisition routines. The field will display “Routine Note Saved!!!” if the current conditions have not been recalled from a stored routine file. The most recently used stored routines are shown as choices in the panel.

Change: If the desired routine is not displayed here, clicking on the **Change** button will allow you to browse through your directories and recall the desired routine.

Save: Stores the currently displayed acquisition parameters and hardware status such as grating used and slit widths in a new routine file.

Save As: Click on this button if you have recalled a routine, modified it and want to preserve the original routine and save the modified routine.

View: Opens a window that shows all of the instrumental parameter settings used when the routine was stored.

Settings panel

Match Hardware: You may want to reproduce the acquisition parameters defined in a routine but not necessarily all of the instrumental parameters. Hardware parameters that can be matched or ignored include the grating choice and slit width settings. If the Match Hardware box is checked, any differences between the current setting and stored settings will be noted in a pop up dialog box.

The screenshot shows the 'Acquisition Routine' panel. At the top, there is a dropdown menu for 'Acquisition Routine' with 'test mono1.rtn' selected, and a 'Change' button to its right. Below this is a 'Routine' section with 'Save', 'Save As', and 'View' buttons. Underneath these buttons is a 'Settings' checkbox (which is checked) and a 'Match Hardware' checkbox (which is unchecked). The 'Operator' field shows 'User 3'. The 'Comments' field contains the text: 'The sample was heated to 300 degrees C for 2 hours before acquisition.' At the bottom, there is a 'Save data as' field with a 'Change' button, and an 'Export To Microsoft Excel' checkbox (unchecked). The 'Acquire' and 'Cancel' buttons are at the very bottom.

This screenshot shows the 'Acquisition Routine' panel with the 'Acquisition Routine' dropdown menu displaying 'Routine Not Saved !!!'. The 'Change' button is to the right of the dropdown. The 'Routine' section below has 'Save', 'Save As', and 'View' buttons. The 'Settings' checkbox is checked, and the 'Match Hardware' checkbox is unchecked. The 'Operator' field is empty, and the 'Comments' field is empty. The 'Save data as' field and 'Export To Microsoft Excel' checkbox are also present at the bottom, along with 'Acquire' and 'Cancel' buttons.

This is a close-up of the 'Settings' section of the panel. It shows a 'Settings' label followed by a checkbox that is currently unchecked, and the text 'Match Hardware' to the right of the checkbox.

Operator/Comments panel

Operator field: Use this field to store the name of the person who acquired the spectra in the file information header. The dropdown list contains the names entered as default users via the **Acquisition Preferences** menu item.

Comments field: Use this field to store comments or other experimental conditions in the data file.

Save panel

Save data as field: Enter the file name you want to use to store your spectra in this field.

Click on the **Change** button to bring up the

Save Data As... dialog box where you can also select a different file format: arc_data, SPC, or txt.

Export to Microsoft Excel: When checked, the data will also be sent to Excel. The number of Excel spreadsheet files generated is based on the number of scans. If you have five cycles, five Excel files will be created.

Export to Grams/32: Available when the SPC format is selected. When checked, data will be exported to GRAMS/32. Refer to "**Chapter 10: Importing Data into G**" for more information.

Export to Grams 3D: Available when the SPC format is selected and Single File, all Scans is the **Storage** mode. When checked, data will be exported to GRAMS/3D. Refer to "**Chapter 10: Importing Data into G**" for more information.

Acquire: Click on this button to start the acquisition. If you have not specified a file name, a dialog box will appear asking if you want to continue or enter a name. You may stop an acquisition at any time by clicking on the **Cancel** button.

Menu Options

File

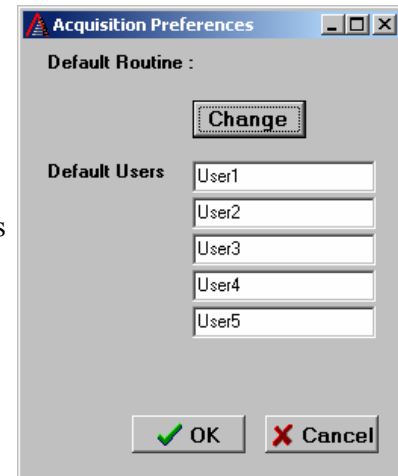
Exit Closes SpectraSense

Select Tab

Hardware Config	Calls up the Hardware Config screen.
Slit Band Pass	Calls up the Monochromator Slit BandPass screen.
Survey Mode	Calls up the Survey mode.
Acquisition	Calls up the Acquisition screen.
Live Data	Calls up the Live Data screen.
Post Processing	Calls up the Post Processing screen.
Mono Control Software	Calls up the Acton Monochromator Control Software screen.
Terminal	Calls up terminal mode for communications debugging. Used for diagnostics in communications between the hardware and computer.

Acquisition Preferences

Preferences... Opens the Acquisition Preferences dialog where you can specify a default routine to be loaded each time that SpectraSense is run. You can also enter the names of up to five default users. These names will be inserted into the Operator dropdown list so the appropriate name can be selected for storage with the data.

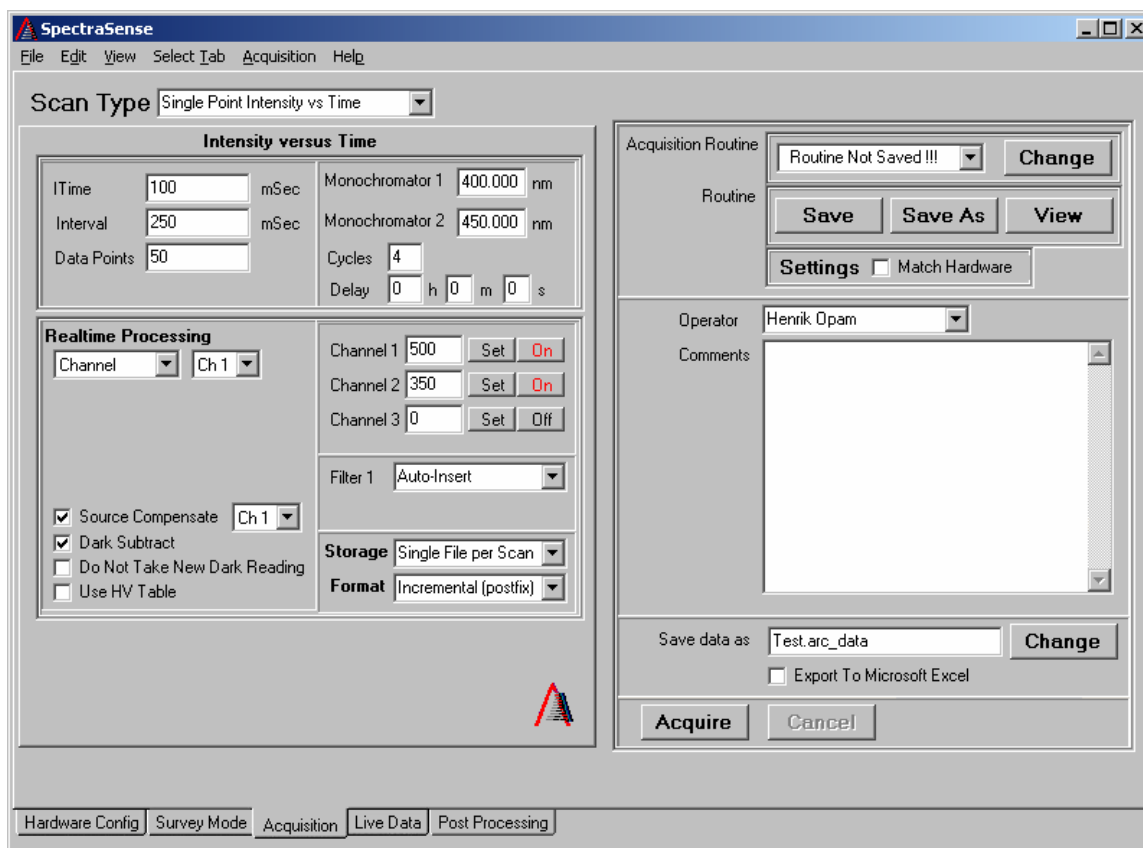


Help

Contents	Hyperlinked index of the SpectraSense manual.
Overview of SpectraSense	Complete SpectraSense Manual on line.
Using the Acquisition Page	Context-sensitive help for this chapter.
SpectraSense Version	The version number of the installed software.

This page intentionally left blank.

Chapter 7: Scanning Time-Based Acquisition



Single Point Intensity versus Time Acquisition

The Single Point Intensity versus Time scanning mode is used to monitor the intensity at a specific wavelength as it changes over time. The spectra created are in intensity versus time units. This mode is valid for measuring intensity changes in intervals as short as 5 milliseconds up to hours.

Note: This mode is not available when an SR-510 is included in the hardware configuration.

Acquisition Parameters

Time panel

The time panel is where the signal integration time, interval between measurements, and the number of data points are specified.

ITime: Integration time is the only parameter passed from the Survey screen. You may change this by highlighting the value in the field and entering a new one.

Interval: This is the time between measurements. The interval must be equal to or larger than the integration time and includes the integration time.

Data Points: This field is used for specifying how many data point are to be taken in the spectrum. If you know how long you want to make the measurements, divide that time by the interval and enter the value in the Data Points field.

A screenshot of the Time panel interface. It contains three input fields: 'ITime' with the value '20' and unit 'mSec', 'Interval' with the value '100' and unit 'mSec', and 'Data Points' with the value '200'.

Wavelength panel

In the wavelength panel you specify the wavelength at which the measurement will be made.

Position: If your system includes two monochromators, two position fields will appear.

Cycles: If you want to make the measurement more than one time, you can specify a number of cycles, where each cycle produces another time base spectrum.

Delay: You can specify a delay between cycles by entering values in the hours, minutes, and seconds fields.

A screenshot of the Wavelength panel interface. It contains four input fields: 'Monochromator 1' with the value '450' and unit 'nm', 'Monochromator 2' with the value '280.000' and unit 'nm', 'Cycles' with the value '1', and 'Delay' with values '0' h, '5' m, and '0' s.

Real Time Processing panel

The Real Time Processing panel normally has values in all of the fields and the appropriate check boxes will be checked. These values are carried over from the Survey Screen or reloaded from a stored routine. You can however change any of the values or check box options.

The Real Time Processing panel is used to define which detectors are to be read out for display and how the incoming data is to be processed before it is displayed.

Display Options: The top left choice field contains all of the display options available, which will vary depending on your system configuration. The choices are Channel and All Channels.

Channel Number: The channel number is chosen from the field to the right of the Display Options field.

Source Compensate: When checked and a source compensation accessory is included in the system, each data point will be divided by the signal from this accessory.

Dark Subtract: When checked, dark subtraction is turned on. A new dark signal will be subtracted from each reading. This signal can be subtracted from all readings when the detector is making a measurement. The software will close any automated shutter and take an initial reading

A screenshot of the Real Time Processing panel interface. It features a 'Channel' dropdown menu set to 'Ch 1'. Below it are four checked checkboxes: 'Source Compensate' (with a 'Ch 1' dropdown), 'Dark Subtract', 'Do Not Take New Dark Reading', and 'Use HV Table'.

or, if the last position on the filter wheel has been specified for dark subtraction, the wheel will move to this position and take a reading and then return to the appropriate position for the measurement. If your system has neither a shutter or filter wheel, you will be requested to block the light from entering the monochromator by any means of your choice, the dark reading will be taken and you will be asked to unblock the light before the actual scan is taken.

Do Not Take New Dark Reading: When checked, a new dark reading will not be taken each time data is acquired ----the most recently acquired dark reading will be used until you uncheck the box.

Use HV Table: When checked, the current HV Table is used to automatically adjust the high voltage on the photomultiplier tube as the monochromator is scanned. Before using this option, make sure that there is an HV table. An HV table can be created or loaded on the Survey Mode Det HV tab.

High Voltage panel

The High Voltage panel will appear if there are photomultiplier tubes in your system. The current value is what was last used in the Survey mode. You may change the high voltage by highlighting the field and entering a new value. Click on the Set button to validate the change. If a stored routine is loaded the high voltage will be changed to the value stored in the routine. You can turn the high voltage on or off by clicking on the On/Off button.

Channel 1	500	Set	On
Channel 2	350	Set	On
Channel 3	0	Set	Off

Filter Wheel panel

The filter wheel panel appears if there is a filter in your system. The filter number or auto insertion option will appear in the choice field. You may change the filter by clicking on the arrow and highlighting the desired filter.

Filter 1	Auto Insert
----------	-------------

Format/Storage panel

Storage field: Active when you are saving the data to an .SPC file. The Storage field allows you to choose either **Single File per Scan** or **Single File, All Scans**.

Storage	Single File per Scan
Format	Average Cycles

Single File per Scan: Saves the data from each cycle into a separate file. If you have three cycles, three files will be created.

Single File, All Scans: Saves the data from each cycle into the same file. If you have three cycles, the generated file will contain three spectra.

Format field: Allows you to specify whether files from the same acquisition program are specified with a numerical prefix, a numerical suffix, average cycles, or a time stamp suffix. If you enter a base filename of "myfile", select the ARC data format, and select Average Cycles, the average cycles option will produce a single data file named myfile.arc_data that contains the averaged data from all of the cycles. Using the same base filename, examples of the other format filenames are:

Numerical Prefix: 001myfile.arc_data - ###myfile.arc_data

Numerical Suffix: myfile001.arc_data - myfile###.arc_data

Time Stamp Suffix: myfile_001_5-9-2006_2h41m22s_pm.arc_data

File Parameters

The File parameters panel is used to save all of the parameters entered in the Acquisition panel into a file (*.rtn) that can be recalled at any time to collect spectra under identical instrumental conditions. Routines can also be recalled in this panel to override the current hardware and operating parameters with those defined in the stored routine file. Although they are not present in this screen, the grating choice and slit width settings used in the original acquisition can also be recalled and set. Data can be stored to a file in either arc_data, SPC, or txt formats or directly exported into an Excel spreadsheet

Routine panel

The Acquisition Routine panel is used to recall or create new acquisition routines. The field will display “Routine Note Saved!!!” if the current conditions have not been recalled from a stored routine file. The most recently used stored routines are shown as choices in the panel.

Change: If the desired routine is not displayed here, clicking on the **Change** button will allow you to browse through your directories and recall the desired routine.

Save: Stores the currently displayed acquisition parameters and hardware status such as grating used and slit widths in a new routine file.

Save As: Click on this button if you have recalled a routine, modified it and want to preserve the original routine and save the modified routine.

View: Opens a window that shows all of the instrumental parameter settings used when the routine was stored.

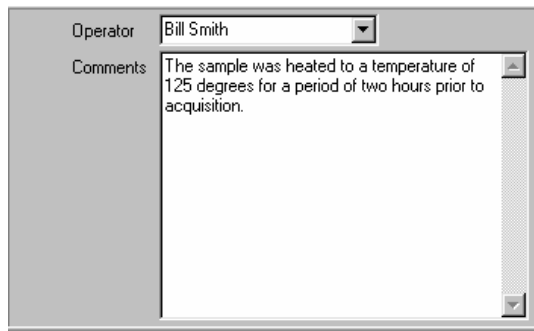
Settings panel

Match Hardware: You may want to reproduce the acquisition parameters defined in a routine but not necessarily all of the instrumental parameters. Hardware parameters that can be matched or ignored include the grating choice and slit width settings. If the Match Hardware box is checked, any differences between the current setting and stored settings will be noted in a pop up dialog box.

Operator/Comments panel

Operator field: Use this field to store the name of the person who acquired the spectra in the file information header. The dropdown list contains the names entered as default users via the **Acquisition Parameters** menu item.

Comments field: Use this field to store comments or other experimental conditions in the data file.

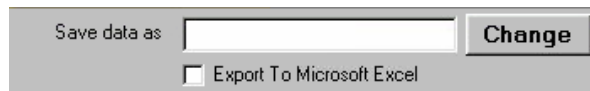
A screenshot of the Operator/Comments panel. It features a dropdown menu for the 'Operator' field, currently showing 'Bill Smith'. Below it is a large text area for 'Comments' containing the text: 'The sample was heated to a temperature of 125 degrees for a period of two hours prior to acquisition.'

Save panel

Save data as field: Enter the file name you want to use to store your spectra in this field.

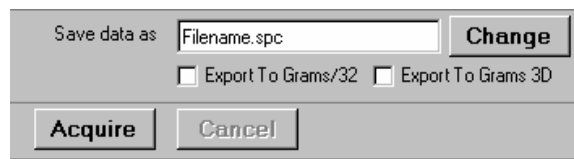
Click on the **Change** button to bring up the

Save Data As... dialog box where you can also select a different file format: arc_data, SPC, or txt.

A screenshot of the Save panel. It shows a text field for 'Save data as' with an empty input box. To the right is a 'Change' button. Below the text field is a checkbox labeled 'Export To Microsoft Excel' which is currently unchecked.

Export to Microsoft Excel: When checked, the data will also be sent to Excel. The number of Excel spreadsheet files generated is based on the number cycles. If you have four cycles, four Excel files will be created.

Export to Grams/32: Available when the SPC format is selected. When checked, data will be exported to GRAMS/32. Refer to "**Chapter 10: Importing Data into G**" for more information.

A screenshot of the Save panel showing additional options. The 'Save data as' field contains 'Filename.spc'. To the right is a 'Change' button. Below the field are two checkboxes: 'Export To Grams/32' and 'Export To Grams 3D', both of which are unchecked. At the bottom are 'Acquire' and 'Cancel' buttons.

Export to Grams 3D: Available when the SPC format is selected and Single File, All Scans is the **Storage** mode. When checked, data will be exported to GRAMS/3D. Refer to "**Chapter 10: Importing Data into G**" for more information.

Acquire: Click on this button to start the acquisition. If you have not specified a file name, a dialog box will appear asking if you want to continue or enter a name. You may stop an acquisition at any time by clicking on the **Cancel** button.

A screenshot showing two buttons: 'Acquire' and 'Cancel'.

Menu Options

File

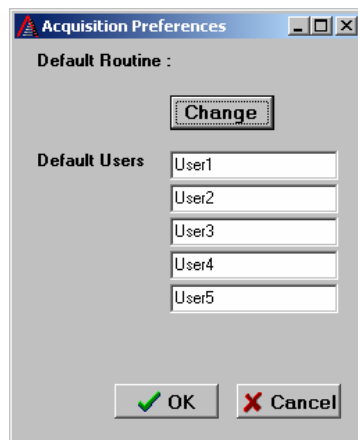
Exit Closes SpectraSense

Select Tab

Hardware Config	Calls up the Hardware Config screen.
Slit Band Pass	Calls up the Monochromator Slit BandPass screen.
Survey Mode	Calls up the Survey mode.
Acquisition	Calls up the Acquisition screen.
Live Data	Calls up the Live Data screen.
Post Processing	Calls up the Post Processing screen.
Mono Control Software	Calls up the Acton Monochromator Control Software screen.
Terminal	Calls up terminal mode for communications debugging. Used for diagnostics in communications between the hardware and computer.

Acquisition Preferences

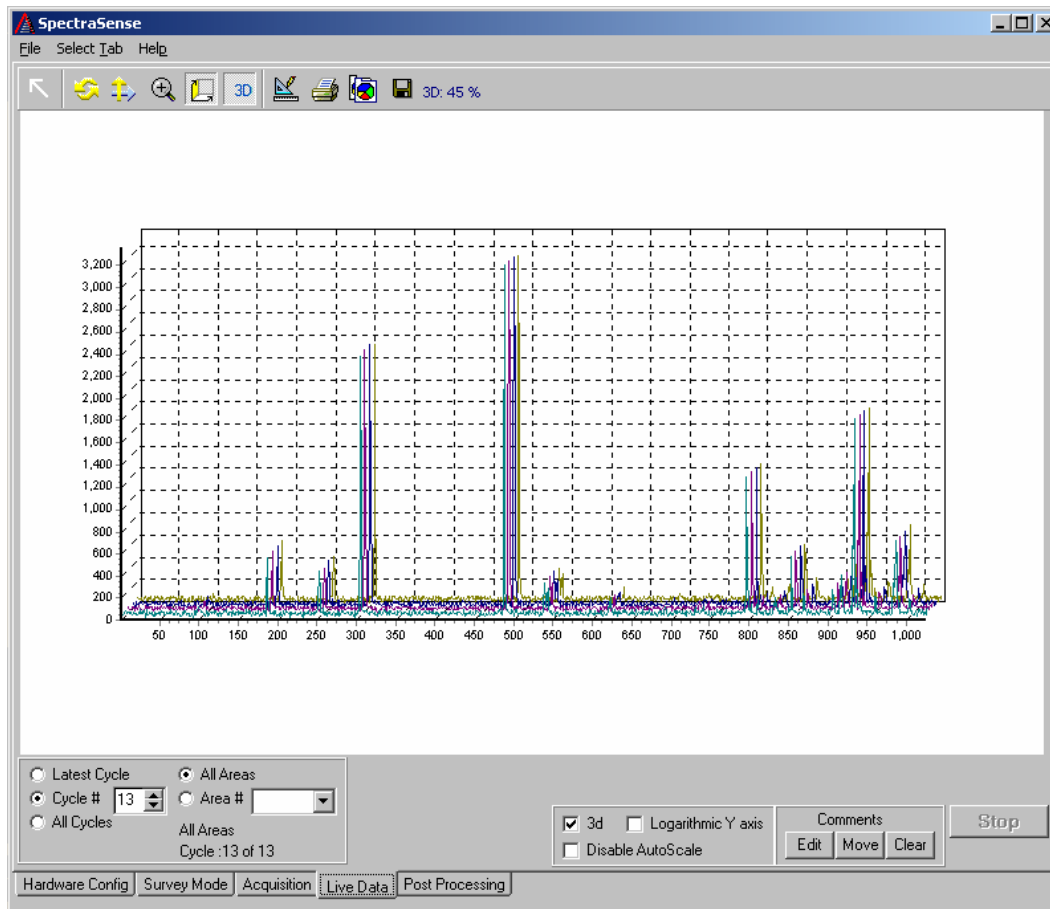
Preferences... Opens the Acquisition Preferences dialog where you can specify a default routine to be loaded each time that SpectraSense is run. You can also enter the names of up to five default users. These names will be inserted into the Operator dropdown list so the appropriate name can be selected for storage with the data.



Help

Contents	Hyperlinked index of the SpectraSense manual.
Overview of SpectraSense	Complete SpectraSense Manual on line.
Using the Acquisition Page	Context-sensitive help for this chapter.
SpectraSense Version	The version number of the installed software.

Chapter 8: Live Data Screen



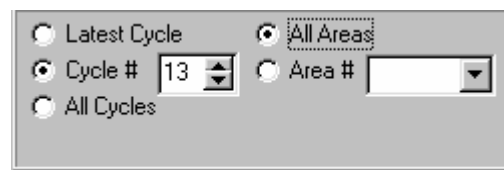
Live Data Display

The live data screen is used to view spectral data as it is being acquired. The screen will update in near real time in all cases except for high speed CCD spectral acquisition.

CCD Mode Display Options

Cycles/Areas

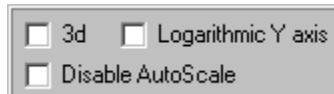
When acquiring spectra it is possible to display the data in different ways. If there were multiple cycles specified in the acquisition it is possible to view at any time, the latest cycle (spectrum) acquired, or the Nth previous cycle (spectrum) individually or all of the currently acquired cycles.



It is also possible, if multiple areas of the chip were read out, to view the spectra from a specific area by selecting it from the Area # dropdown list or to view all of the areas' spectra simultaneously.

Scaling

Data can be displayed using a linear or logarithmic scale. It can be displayed with multiple spectra overlaid on XY axes or with a third axis. By default the data is autoscaled to fit the screen, but autoscaling can be turned off.



A panel with three checkboxes: ☐ 3d, ☐ Logarithmic Y axis, and ☐ Disable AutoScale.

3d: When checked, multiple spectra will be displaced so that the most recently collected spectrum appears in front of the previous ones. If unchecked, the spectra will be overlaid.

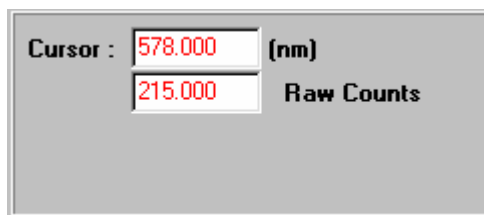
Logarithmic: When checked, the data can be presented on a logarithmic scale for data with a very wide dynamic range. If unchecked, the data is drawn using a linear scale.

Disable AutoScale: When checked, autoscaling is turned off. If unchecked, autoscaling is active.

Scanning Mode Display Options

Cursor panel

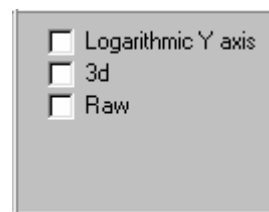
The cursor panel will display the current position and intensity values for all active detectors.



A panel showing cursor data: 'Cursor : 578.000 (nm)' and '215.000 Raw Counts'.

Scaling

Data can be displayed using a linear or logarithmic scale. It can be displayed with multiple spectra overlaid on XY axes or with a third axis. Typically, the displayed data is the result of any mathematical operations included in the acquisition setup. However, the raw data can be displayed instead.



A panel with three checkboxes: ☐ Logarithmic Y axis, ☐ 3d, and ☐ Raw.

Logarithmic Y axis: When checked, the Y axis will be logarithmically scaled. If unchecked, linear scaling will be used.

3d: When checked, multiple spectra will be displaced so that the most recently collected spectrum appears in front of the previous ones. If unchecked, the spectra will be overlaid.

Raw: When checked, the raw intensity values for all detectors in use will be displayed. This option is generally used for debugging unexpected results by verifying that all the detectors are functioning and are in the light path. If unchecked, the specified real time display may be the resultant of channel1/channel2 or a channel divided by a stored data file.

Stop Button

The stop button will stop an acquisition in progress. All valid data collected up until the button was clicked on will be saved in the data file.



A button labeled 'Stop'.

Header/Comment

You can annotate the plot by adding a Header and a Comment by using the buttons in the Comments panel.

Comments panel

Edit: Opens the Comment Form where you can enter or change a header and/or a comment.



Move: This button allows you to re-position a comment on the plot. After clicking on this button, depress the left mouse key and drag the comment to its new location. Release the mouse key and then click on the **Done** button.

Clear: Deletes the comment and the header from the plot. Note that the header text will still appear in the Comment Form but it will not print out.

Comment Form dialog

Header: By default, the plot header will either be blank or will be TChart. To use your own header, enter the text you want to appear on the plot and use the **Font Size** up/down arrows to change the character size. Click on **OK** to apply the header and end the editing session. The header text will print on the plot when you use the File menu **Print Spectra** function. This header will be applied to all subsequent plot printouts until you edit or delete it in the Comment Form dialog or until you close SpectraSense.

Comment: Enter the comment you want to appear on the plot. Use the Font Size up/down arrows to change the character size. When you click on **OK**, the comment will be positioned in the lower left corner of the data plot. If you prefer in another location, click on the **Move** button in the Comments panel. This comment will be applied to all subsequent plot printouts until you edit or delete it in the Comment Form dialog or until you close SpectraSense.



Plot Formatting Tools



This button is informational. It tells you how to expand and move the **data** within the plot area of the screen. Click and drag with the left mouse button for expansion and with the right for positioning



These buttons affect the size and positioning of the plot, including the axis on the screen. The left icon will rotate the plot. The second will move the whole plot on the background. The magnifying glass will expand and shrink the plot on the background. The last icon will adjust the spacing of spectra in 3D plots.



These buttons are used to change the presentation of the plots. The left button will open a complete dialog box of features for adjusting everything from plot color to adding legends and annotation. The second button will print the screen while the third button will copy the existing plot to the Windows clipboard. The fourth button allows you to save the plot as to a TEE, BMP, EMF, or WMF file.

Menu Options

File

Copy Chart to Clipboard	Copies the image displayed or the spectrum or spectra displayed to the Windows clipboard.
Print Frame	Sends a bitmap image of the screen to the printer.
Spectra	Prints a date-time stamped plot of the spectrum or spectra displayed. If the data was saved to a file, the directory location will also be included on the plot.
Print Setup	Calls Windows printer control screen.
Exit	Closes SpectraSense.

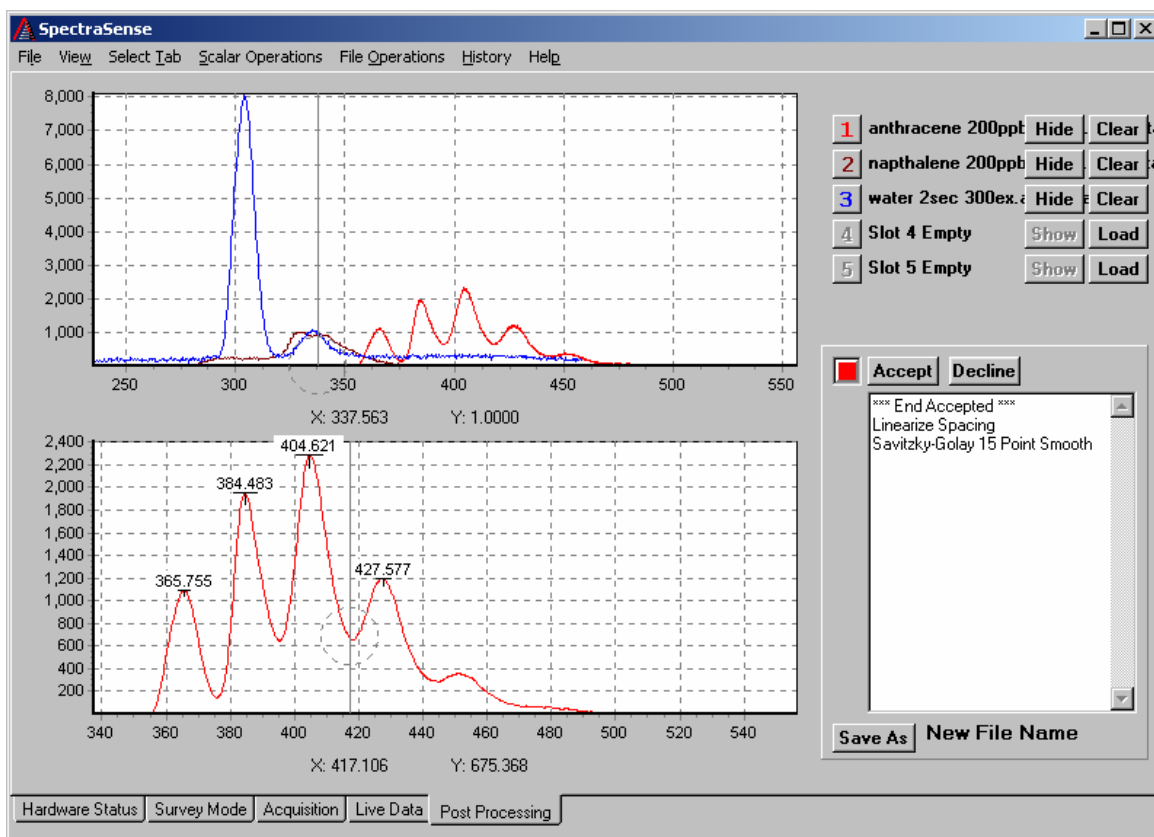
Select Tab

Hardware Config	Calls up the Hardware Config screen.
Slit Band Pass	Calls up the Monochromator Slit BandPass screen.
Survey Mode	Calls up the Survey mode.
Acquisition	Calls up the Acquisition screen.
Live Data	Calls up the Live Data screen.
Post Processing	Calls up the Post Processing screen.
Mono Control Software	Calls up the Acton Monochromator Control Software screen.
Terminal	Calls up terminal mode for communications debugging. Used for diagnostics in communications between the hardware and computer.

Help

Contents	Hyperlinked index of the SpectraSense manual.
Overview of SpectraSense	Complete SpectraSense Manual on line.
Using the Live Data Page	Context-sensitive help for this chapter.
SpectraSense Version	The version number of the installed software.

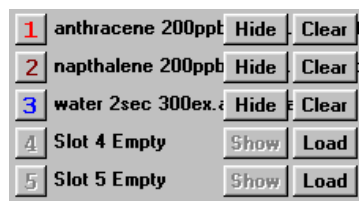
Chapter 9: Post Processing Screen



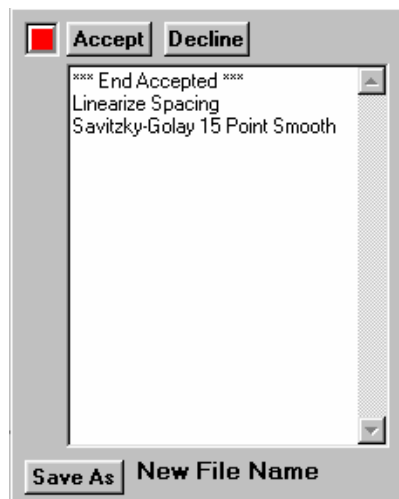
The Post Processing screen is used for evaluation and treatment of your spectral files. Up to five files may be viewed simultaneously. The Post Processing screen has a full complement of classically used mathematical manipulations employed in spectral analysis.

The Post Processing screen has two display windows. The top window shows the unaltered spectra. The bottom shows the resultant spectrum after it has been operated on. The spectra in each window can be independently expanded or contracted. A cursor in each window displays the position and intensity information at its location.

Up to five spectra can be loaded for display in the top window. Once loaded, a spectrum can be shown or hidden from view by clicking on the Show/Hide button. A spectrum can be cleared or loaded by clicking on the Load/Clear button. The numbered buttons to the left of the spectrum's name are used to select which spectrum the cursor will track and the spectrum to be manipulated. The chosen spectrum is also displayed in the bottom window.



The manipulations log panel keeps track of every operation performed on the spectrum. Each operation or series of operations can be accepted or declined by clicking on the appropriate button. If you want to save the treated spectrum, click on the Save As button. The save file as dialog box will appear. All of the operations performed on the spectrum and the date they were performed are saved in the new data file's header.



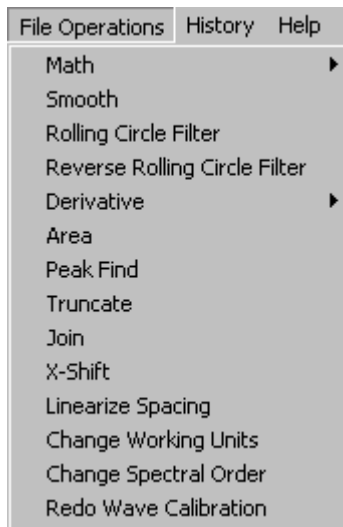
Mathematical Operations

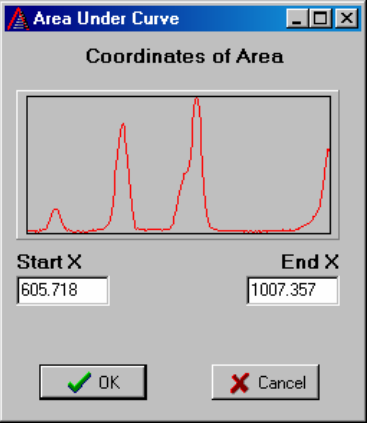
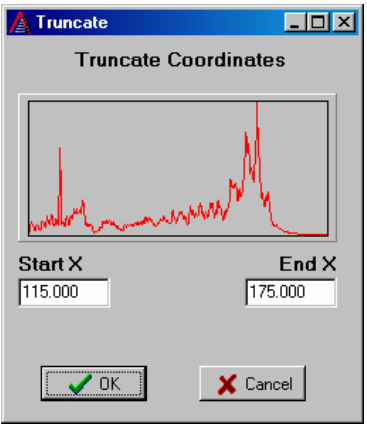
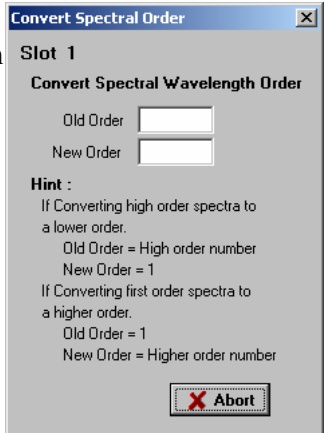
Scalar Operations

Scalar	Data + - * / k , k-data, k/Data
Log	
Ln	
Antilog	Antilog e, Antilog 10

File Operations

Math	addition, subtraction, division, multiplication with another file
Smooth	Savitzky-Golay 5, 9, 15, points
Rolling Circle Filter	Subtracts a background (fluorescence or offset) by using the entered radius and number of iterations. Useful when the signal of interest is small in comparison to the background
Reverse Rolling Circle Filter	Performs a peak removal and leaves the background. Useful for removing peaks and cosmic rays.
Derivative	First, Second



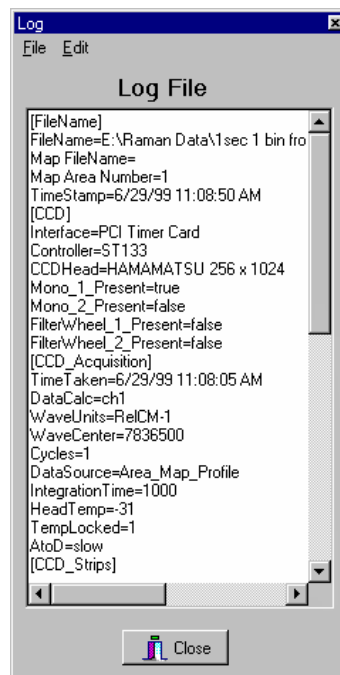
Area	Allows you to find the area under a peak. The Coordinates of Area dialog appears when area is selected from the File Operations Menu. Enter the Start X and End X values into the fields. When you click on Ok , the baseline and total area for the specified range will be calculated and displayed. No change is made to the data.	 The 'Area Under Curve' dialog box shows a plot of a spectrum with two distinct peaks. Below the plot, the 'Start X' field is set to 605.718 and the 'End X' field is set to 1007.357. There are 'OK' and 'Cancel' buttons at the bottom.
Peak Find	Finds the highest peak in the spectrum gives coordinates and half width.	
Truncate	Allows you to discard data points from ends of spectra. The Truncate dialog appears when truncate is selected from the File Operations Menu. Enter the specific Start X and End X values into the fields. The operation will occur when you click on Ok and the lower window will be updated with the truncated data.	 The 'Truncate' dialog box shows a plot of a spectrum. Below the plot, the 'Start X' field is set to 115.000 and the 'End X' field is set to 175.000. There are 'OK' and 'Cancel' buttons at the bottom.
Join	Attaches another spectrum to the end of the selected spectrum	
X shift	Changes the wavelength calibration of the spectrum by shifting the x-axis coordinates by the shift value.	
Linearize	Specific to CCD spectra. Extrapolates the data such that the data point spacing is in equal wavelength units.	
Change Working Units	Converts calibration between all supported working units in the software; nm, Å, microns, eV, absolute and relative wavenumbers.	
Change Spectral Order	Opens the Convert Spectral Order where you can change the wavelength units of a spectra to a different order. This allows you to see what the current spectral set would look like if it was a different order. For example, if you have second order data in your spectra, you could see what the data looks like in first order. This is a fringe operation.	 The 'Convert Spectral Order' dialog box is for 'Slot 1'. It has a section 'Convert Spectral Wavelength Order' with 'Old Order' and 'New Order' input fields. Below this is a 'Hint' section with text explaining how to convert between orders. At the bottom is an 'Abort' button.

Redo Wave Calibration Allows you to replace the current wavelength calibration with a calibration that comes from an ARC Wavelength Data file (*.arc_wave) or an ARC Wavelength Data (ini file) file (*.ini). The files are generated by outside tools. This function mainly applies to the InSpectrum 150 and the HTS Lens Spectrograph.

History

- Show Pair List** Opens a dialog box with the X Y coordinates of the data. The data may be printed out or copies to the clipboard.
- Show Peak List** Show all found peaks at the specified discrimination level. The data may be printed out
- Discrimination** Used to define peaks to be labeled. Enter a value between 1 and 100 percent of full scale. All peaks with heights above that level will be marked.
- Show/Hide Peaks** Labels or clears peak position labels on the lower display screen.

Show Log File Opens a dialog box with all of the stored hardware, acquisition, and data treatment parameters. The Log File contains all of the information on how and when the data file was acquired. This information can be printed out from the File menu. The log file can be saved as an individual file without the data by selecting Save from the File menu. Data can be cut and pasted from the Edit menu level. There is no way to modify the information directly within SpectraSense. If you want to do so, you may open the data file in a text editing or word processing software package.



Recalculate Components If a spectrum is acquired where there are two detectors used in the acquisition such as CH1/CH2, the raw data from each of the detectors is also stored in the file. Although only the resultant spectrum is normally displayed, the raw data is always available for diagnostic and reconstructive purposes. The Recalculate Components option allows you to perform mathematical procedures on the individual components and to reconstitute the derived spectrum from the raw data.

Show/Hide Components Allows you to switch between seeing the spectrum as defined in the real time processing or to see the raw input from each of the detectors.

Menu Options

File

Load	Loads a data file into a slot
Save As	Saves a data file under a new name
Copy to Clipboard	Saves the selected chart (Top or Bottom) to Clipboard
Export to Excel	Exports the data displayed in the bottom window to an Excel spreadsheet in two column format. Column A contains the X coordinates and Column B contains the Y.
Wave Calculator	Spectral units converter
Wavenumber Calculator	Calculates spectral positions for relative wavenumber units.
Clear	Clears a data slot
Delete	Deletes a data file
Print Spectrum	Prints a plot of the spectrum or spectra displayed. You can choose to print the plot in the top window or in the bottom window.
Screen	Prints an image of the SpectraSense screen.
Data	Prints out the entire data file including parameters and data coordinates.
History	Prints the log file from the data set. The log file lists all of the stored hardware, acquisition, and data treatment parameters stored with the data.
Peak list	Prints all found peaks at the specified discrimination level.
Exit	Closes SpectraSense.

View

Autoscale	Automatically scales the axes of the plot so that the full spectrum is shown.
Full Width	Show the spectrum with the current vertical expansion but full wavelength coverage.
Full Height	Shows the spectrum with full vertical expansion but with current horizontal expansion.
Fixed Expansion	Pops up the Zoom dialog where you can key in the start and end X and Y coordinates for the charts. After entering the coordinates, click on OK. The dialog will close and the charts will be redrawn.
Zoom In	Expands the spectrum to show half the current values as full scale on the display.
Zoom Out	Shrinks the current expansion to twice the current level.

Select Tab

Hardware Config	Calls up the Hardware Config screen.
Slit Band Pass	Calls up the Monochromator Slit BandPass screen.
Survey Mode	Calls up the Survey mode.
Acquisition	Calls up the Acquisition screen.
Live Data	Calls up the Live Data screen.
Post Processing	Calls up the Post Processing screen.
Mono Control Software	Calls up the Acton Monochromator Control Software screen.
Terminal	Calls up terminal mode for communications debugging. Used for diagnostics in communications between the hardware and computer.

Scalar Operations : See page 76

File Operations: See page 76

History: See page 78

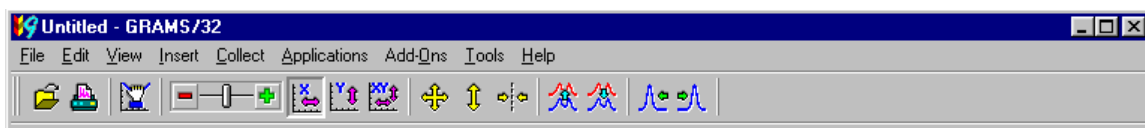
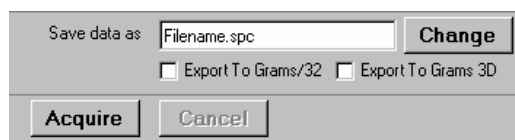
Help

Contents	Hyperlinked index of the SpectraSense manual.
Overview of SpectraSense	Complete SpectraSense Manual on line.
Using the Post Processing Page	Context-sensitive help for this chapter.
SpectraSense Version	The version number of the installed software.

Chapter 10: Importing Data into GRAMS®

SpectraSense software can automatically transfer data into Galactic Industries' GRAMS/5 and all of the related programs such as GRAMS/3D and PLSplus/IQ. There are two methods of working with GRAMS. The first method is to run the two programs concurrently and let SpectraSense send the data to GRAMS via DDE, direct data exchange. This is accomplished in the Survey Mode by clicking on the Edit menu level. You will have the option to export the data to GRAMS or GRAMS/3D when using a CCD or only GRAMS when working with a single channel detector.

In the Acquisition screen, if you specify a filename using the .SPC format in the Save dialog box, two check boxes will appear. They are **Export to Grams** and **Export to Grams 3D**. If one of these boxes is checked, the data will be passed to the program and will appear in the working window. This method has the advantage, when taking long acquisitions, of allowing you to continue to work with previously collected data in the GRAMS window.



The second method of transferring data directly into GRAMS is to use the Collect option on the menu bar in GRAMS. When you select Survey Mode or Scan, a window will open with the selected SpectraSense operation. You can then input the parameters necessary for the acquisition and initiate it. When the acquisition is complete you will be returned to the main GRAMS screen. If you use this method you can not change to another GRAMS window or work on another data file until the acquisition is completed.

This page intentionally left blank.

Glossary

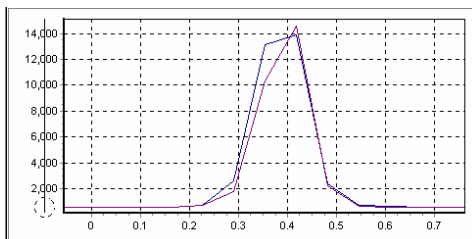
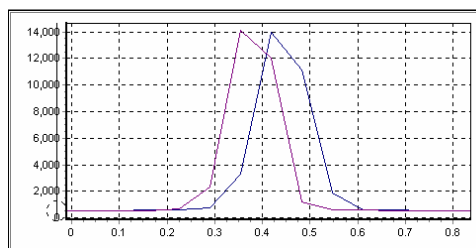
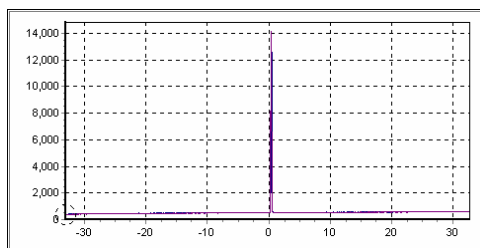
- Acquisition** A spectral acquisition is the displayed or stored results of a single readout or group of readouts. A spectral acquisition may generate several spectral data files or a multi-file with several spectra in it depending on the parameters selected.
- Area** An area is a user-defined portion of the chip from which an image or spectrum will be acquired. Multiple areas can be created on the chip and information from each area can be stored as separate data sets as either spectra or images as desired. Areas may not overlap. In spectral mode, the data from each area may be stored as an individual data file or as a subfile in a multi-file.
- Cleaning** Cleaning is the continuous readout of the CCD without digitization between acquisitions. This eliminates the accumulation of dark charge in each readout of interest.
- Cosmic Correction** Cosmic rays are always traveling through the atmosphere. When high energy cosmic rays strike the CCD chip they release electrons in a manner similar to photons. The result is spurious peaks. To differentiate between photon induced peaks and cosmic ray peaks, two spectra are taken under the identical conditions and are compared. If there is a peak in one spectrum and not in the other, it is assumed to be produced by a cosmic ray and the resultant spectrum, which is saved, does not show this peak.
- Dark subtraction** Electrons generated from thermal processes rather than interactions with photons are always present on each readout. The rate at which they are generated is constant for any given temperature. By taking a readout with the shutter closed and subtracting the data from a readout of the same integration time with the shutter open, you can subtract out the thermionic contributions to the data. In most acquisitions, when dark subtraction is selected, a single readout of the array is made first with the shutter closed.
- Group** A group is a user-selected set of contiguous columns in an area in which all the data will be added into one intensity value. A group is only applicable in Peak Monitoring mode and is used in comparing the integrated intensity under one peak with another.
- Horizontal binning** Horizontal binning is the combining of adjacent columns into one wider column. Horizontal binning is valid in both spectral and image modes. In image mode it reduces horizontal spatial resolution but increases the intensity of each superpixel. In spectral mode horizontal binning will increase the signal to noise ratio by the square root of the binning factor but will reduce the measurable spectral resolution inversely to the binning factor. In low light applications where the spectral features are broad compared to the pixel size, it would be advantageous to employ horizontal binning.

Image	An image is a three-dimensional dataset where each data point has an intensity value, a horizontal pixel position, and a vertical pixel position. Images are used mainly for diagnostic and setup purposes in SpectraSense CCD software.
Multi-file	When data is stored in GRAMS®-compatible format, multiple spectral readouts can be stored in a single data file. These readouts may be either the spectra from multiple areas on the chip or spectra taken sequentially with a time delay.
Readout	A readout is the result of digitizing the data from the chip. A readout can result in a spectrum, a subset of a spectrum, if a final spectrum is to be made from the addition of several readouts, or an image. It can also produce an image.
Segment	A segment is part of a spectral acquisition taken with the range option. Multiple segments are “glued” together to construct a large spectral coverage spectrum.
Software binning	Software binning is the coaddition of the intensity values from each row in the selected area of the chip. In this case, each row is individually sent through the chip’s readout register and digitized rather than filling the readout register with all of the charge from every row in the area before digitizing. It is only applicable in spectral mode. Software binning can prevent saturation of intense spectral features while improving the signal to noise ratio on weaker features. The penalty for using this option is that the time to read out a spectrum is significantly increased. To read out a full 1024 x 256 array using software binning can take about 3 seconds. Without spectral binning it takes milliseconds.
Spectrum	A spectrum is a two-dimensional data set where each point contains an intensity value and a position value or a time value. The positional value can be stored in terms of dispersed wavelength or horizontal pixel number on the chip.
Vertical binning	Vertical binning is the co-addition of adjacent rows into a single value. In image mode, it will reduce the vertical spatial resolution of the image with a linear increase in intensity. In spectral mode, all of the rows are combined automatically by the hardware in the readout process, therefore vertical binning is valid parameter.

Appendix A

Focusing Your Princeton Instruments CCD Detector mounted on an Acton Spectrometer

1. Attach the CCD detector mounting flange to the face of your CCD detector head. Slide the adapter into the axial exit port of the spectrograph and align the head so that the long axis of the chip is parallel to the table-top. Make all of the connections between the detector, shutter, and computer. Turn on the detector and the computer. Run SpectraSense and go to the Survey screen. Wait for the detector to reach operating temperature.
2. Illuminate the entrance slit of the spectrograph with an Hg pen lamp or light from an overhead fluorescent lamp. Open the entrance slit to approximately 50μ . If your spectrograph has more than one entrance slit, be sure that you are illuminating the slit with the CCD detector shutter and that the turning mirror is in the correct position. (Verify this by looking at the optical path in the Hardware Config screen).
3. Set the monochromator position to 546 nm. Set the integration time to approximately 250 ms. Click on the Spectral/Imaging button so that the display is in spectral mode. Press the **GO** button. You should see a peak in the center of the display. You may have to adjust the integration time so that it is clearly visible and not oversaturated. Click on **Stop**.
4. Click on the **Area Maps** tab. On the Quick Strips panel, enter 4 in the field and click on the Create button. This will create 4 equal areas on the chip. Click on the **Hide** buttons for areas 1 and 4. This will leave you with two areas in the center of the chip.
5. Click on the **Setup** tab and then click on **GO**. Adjust the integration time so that the two 546 nm peaks are clearly visible. Expand the display around the peak so that you can easily see how their shapes change. Slide the adapter in and out slowly until the peaks are at their maximum height and minimum width. Next, slowly rotate the detector so that the two peaks are superimposed. Once you are satisfied with the adjustment, lock the adapter in place using the two locking screws located on the top and side of the spectrograph.



This page intentionally left blank.

Appendix B

Improving the signal to noise ratio of your CCD spectra

The name of the game in acquiring spectra is to get a good signal to noise ratio (s/n) so that you can say with certainty that a feature is a true spectral peak and not just a randomly higher noise spike. There is no fixed value that must be attained in order to say that a spectrum is good or noisy. For strong phenomena it is not uncommon to have a s/n of over 1000. In certain very weak Raman and fluorescence experiments, attaining a s/n of 3 or 5 may be considered to be sufficient.

There are several factors that contribute to the noise component of a spectrum. They include the random component of the dark signal generated in the detector, the random component of the signal being measured, and the random signal added by the electronic circuitry. The random components of the dark signal and analytical signals being measured are the square root of the signals. For a signal of 400 counts the noise would be 20 counts thereby giving a s/n of 20. If a detector has an intrinsic dark signal of 2500 counts/sec and the signal being measured is 30 counts/sec, it would not be possible to detect the analytical signal because, even if we could subtract out the dark signal, its random component would be 50 counts/sec, which is greater than the signal we are trying to measure.

The advantage of using CCD detectors is that the random components due to the dark signal and the readout are so low that for most applications the s/n of the spectrum is only limited by its own random component. When this condition is met, the measurement is said to be shot noise limited.

Most scientific CCD detectors are cooled. The colder the CCD chip is, the lower the dark signal will be. A general rule of thumb is that for every 7 degrees C that the CCD is cooled, the dark signal halves. With a liquid nitrogen (LN)-cooled CCD the dark signal can be less than 1 electron/pixel/hour. As there are typically between 5 to 8 electrons per “count”, it would therefore take about 8 hours to produce 1 count with such a detector. Light levels which are much less than 1 photon/sec can be detected with an LN-cooled CCD. A typical thermoelectrically-cooled CCD produces approximately 5 to 8 electrons/pixel/second, which can be roughly translated to 1 count/sec/pixel. The most commonly used CCD chips in spectroscopy are 256 pixels high. If all of the pixels are binned for a data point, there will be 256 dark counts/sec producing a 16 count/sec non-subtractable noise component to any measurement.

It should be clear that to improve the s/n the measured signal level must be increased. All other external parameters aside, there are only a few things that you can do to improve the s/n. They are:

1. Increase the integration time. As all of the electrons being released by the experimental signal are being held before readout, the more collected the more counts will be produced. Doubling the integration time will double the number of counts measured for a steady state experiment. By doubling the integration time you are increasing the s/n by the square root of 2 or 1.4. It is noticeable but not dramatic. A s/n gain of 2 is noticeable.
2. Open the entrance slit of the spectrograph to let more light in. Double the width, you double the signal as long as the illumination spot is wider than the slit width. Increasing the slit width has the adverse effect of reducing the spectral resolution. However, unless

you are looking at very narrow spectral features, you probably can afford to double the slit width without degrading resolution.

3. You can increase the effective size of your detector. You increase the size of your detector by binning pixels together horizontally. This procedure is very effective for broad band phenomena such as fluorescence and absorption. You can increase the horizontal binning factor to the number of pixels it takes to equal your entrance slit width, without degrading spectral resolution. If your entrance slit is 100 microns and your pixels are 25 microns wide, you can bin four pixels horizontally without changing the spectral resolution and increase your s/n by a factor of 2!
4. You can decrease your effective detector size by reading out as few rows as possible assuming the light from your experiment is focused down to a spot with a diameter less than the height of the CCD. This is especially important when measuring weak signals through a fiber. If a fiber is 200 microns in diameter and the chip is 256 x 25 micron pixels high, the dark signal and its inherent noise component from the 248 pixels not being illuminated can visibly degrade the s/n of the spectrum. In this case you would take an image and create an area encompassing only the spectrum from the fiber.
5. In cases where the measurement includes very strong peaks as well as very weak peaks, software binning can be used to increase the dynamic range allowing you to integrate long enough to get a good s/n on the weak peaks without clipping the strong peaks. This is possible because when a spectrum is normally taken, all of the charge from all of the rows in a column is dumped into the readout register of the CCD at one time. It is possible, that perhaps, 50 rows worth of charge might not saturate the readout register but 250 would. By reading each row out individually and coadding the values in software, it is possible to collect the maximum signal for the weak peaks as well for the strong peaks without saturation. The penalty that is paid for this is that the readout noise associated with electronics is added 256 times instead of one. However, since the readout noise is very low, it probably will not degrade the s/n too much from the theoretical gain. A second penalty is that it takes much longer to read out a spectrum because all of the pixels must be individually digitized.
6. If you are not resolution limited, you can select a grating with a coarser groove density. This will reduce the dispersion and increase the "number of wavelengths" on each pixel. Changing from a 1200g/mm grating to a 600g/mm grating with the same blaze characteristics will increase the s/n by 1.41 and double the spectral coverage.

The trick is to use any combination of these techniques to get the highest quality spectra in the shortest amount of time possible. You can see the effect each has in real time by using the adjustments in the Survey Mode screen.

Appendix C

Trigger order of operation

Trigger operations for each of the 6 phases of a cycle occur in four phases

Input triggers that need to be reset are reset

1. Output lines for a complementary phase are reset
2. Output lines that need to be set are set
3. Wait for external I/O for the phase

The order in which events occur during a scan

1. Before a routine starts
 1. Clear and setup all input triggers
 2. Set all Output line initial states
 1. High output trigger lines are set low
 2. Low output trigger lines are set high
 3. Rising edge trigger lines are set low
 4. Falling Edge trigger lines are set high
 3. Set all start routine output line trigger states (1 first, 4 last)
 4. Wait for all start routine input line trigger states or abort trigger states (in order 1 first, 4 last)
 1. Check for the start routine trigger (in order 1 first, 4 last)
 2. Check for any abort scan triggers (in order 1 first, 4 last)
 5. Check for any abort scan input triggers (in order 1 first, 4 last)
 6. If Scan Aborted set any Scan abort output lines (in order 1 first, 4 last)
2. Before a cycle starts
 1. Set all start cycle output line trigger states (1 first, 4 last)
 2. Clear and set all End Cycle output lines to initial states
 1. High output trigger lines are set low
 2. Low output trigger lines are set high
 3. Rising edge trigger lines are set low
 4. Falling Edge trigger lines are set high

3. Wait for all start cycle input line trigger states or abort trigger states (1 first, 4 last)
 1. Check for the start routine trigger (in order 1 to 4)
 2. Check for any abort scan triggers (in order 1 to 4)
4. Check for any abort scan input triggers (in order 1 first, 4 last)
5. If Scan Aborted set any Scan abort output lines (in order 1 first, 4 last)
3. Before a point is taken (if Input Abort or wait for trigger are set, otherwise skipped)
 1. Wait for all each point input line trigger states or abort trigger states (1 first, 4 last)
 1. Check for the start routine trigger (in order 1 first, 4 last)
 2. Check for any abort scan triggers (in order 1 first, 4 last)
 2. Check for any abort scan input triggers (in order 1 first, 4 last)
 3. If Scan Aborted set any Scan abort output lines (in order 1 first, 4 last)
4. After a point is taken (if Input Abort or wait for trigger are set, otherwise skipped)
 1. Clear and setup all each point input trigger lines (1 first, 4 last)
 2. Check for any abort scan input triggers (in order 1 first, 4 last)
 3. If Scan Aborted set any Scan abort output lines (in order 1 first, 4 last)
5. After a cycle completes
 1. Clear and setup all start cycle input trigger lines (1 first, 4 last)
 2. Reset all start cycle output line trigger states (1 first, 4 last)
 3. Set all cycle complete output line trigger states (1 first, 4 last)
 4. Check for any abort scan input triggers (in order 1 first, 4 last)
 5. If Scan Aborted set any Scan abort output lines (in order 1 first, 4 last)
6. After a routine completes
 1. Reset all start routine output line trigger states (1 first, 4 last)
 2. Set all routine complete output line trigger states (1 first, 4 last)
 3. Check for any abort scan input triggers (in order 1 first, 4 last)
 4. If Scan Aborted set any Scan abort output lines (in order 1 first, 4 last)

Appendix D

Image Tiff 6.0 Compliance

This document, documents how SpectraSenseCCD is compliant with the Tiff 6.0 Specification. SpectraSense CCD is capable of storing three different types of tiff files.

1. 8 Bit Gray Scale Images.

Compliant with Section 4 of Tiff 6.0,

Tags Used :

Tag	Type	Count	Values
• 0100 hex	4	1	Image Height
• 0101 hex	4	1	Image Width
• 0102 hex	3	1	8
• 0103 hex	3	1	1, no compression
• 0106 hex	4	1	1, 0 is Black
• 010f hex	2	14 hex	'Acton Research Corp.'
• 0111 hex	4		ImageWidth,Table of Strip offsets
• 0115 hex	3	1	1
• 0116 hex	4	1	1 row per strip
• 0117 hex	4		ImageWidth,ImageWidth
• 011a hex	5	1	Pixel Width (CM)
• 011b hex	5	1	Pixel Height (CM)
• 0128 hex	3	1	3 (CM)
• 0131 hex	2	10 hex	'SpectraSenseCCD'
• 0132 hex	2	20	Tiff 6.0 Time Format

Internal Layout of File

- 8 Byte Tiff Header 49 49 2a 00 08 00 00 00 hex
- IDF Table
- Variables over 4 Bytes Table
 - DateTime Data
 - Make String
 - SoftwareString

- Xres Value
- Yres Value
- Strip Offsets
- Strip Counts
- Data Bytes, in order starting with Strip 0

*Note : All IDF Table offsets are justified on 16-bit words. Image strip offsets are not justified on 16-bit words. An image will have the Next Strip offset 1 StripWidth * bytes past the last strip. All Tiff 6.0 compliant Tiff readers will read this. Some Non-Compliant Tiff readers do NOT.*

2. 8,8,8 Bit RGB False Color Images

Compliant with Section 6 of Tiff 6.0,

Tags Used :

Tag Type	Count	Values
• 0100 hex	4	1 Image Height
• 0101 hex	4	1 Image Width
• 0102 hex	3	3 8,8,8
• 0103 hex	3	1 1, no compression
• 0106 hex	4	1 2 True RGB
• 010f hex	2	14 hex 'Acton Research Corp.'
• 0111 hex	4	ImageWidth, Table of Strip offsets
• 0115 hex	3	1 3
• 0116 hex	4	1 1 row per strip
• 0117 hex	4	ImageWidth, ImageWidth * 3
• 011a hex	5	1 Pixel Width (CM)
• 011b hex	5	1 Pixel Height (CM)
• 011c hex	3	1 1 (chunky)
• 0128 hex	3	1 3 (CM)

Internal Layout of File

- 8 Byte Tiff Header 49 49 2a 00 08 00 00 00 hex
- IDF Table
- Variables over 4 Bytes Table
 - DateTime Data
 - Make String
 - Software String

- Xres Value
- Yres Value
- Bits Per Sample
- Strip Offsets
- Strip Counts
- Data Bytes, in order starting with Strip 0

*Note : All IDF Table offsets are justified on 16-bit words. Image strip offsets are not justified on 16-bit words. An image will have the Next Strip offset 3 Strip Widths * bytes past the last strip. All Tiff 6.0 compliant Tiff readers will read this. Some Non-Compliant Tiff readers do NOT.*

3. 16 Bit Gray Scale Images.

An Extension to Section 4 of Tiff 6.0,

Tags Used :

Tag	Type	Count	Values
• 0100 hex	4	1	Image Height
• 0101 hex	4	1	Image Width
• 0102 hex	3	1	16
• 0103 hex	3	1	1, no compression
• 0106 hex	4	1	1, 0 is Black
• 010f hex	2	14 hex	'Acton Research Corp.'
• 0111 hex	4		ImageWidth, Table of Strip offsets
• 0115 hex	3	1	1
• 0116 hex	4	1	1 row per strip
• 0117 hex	4		ImageWidth, ImageWidth * 2
• 011a hex	5	1	Pixel Width (CM)
• 011b hex	5	1	Pixel Height (CM)
• 0128 hex	3	1	3 (CM)
• 0131 hex	2	10 hex	'SpectraSenseCCD'
• 0132 hex	2	20	Tiff 6.0 Time Format

Internal Layout of File

- 8 Byte Tiff Header 49 49 2a 00 08 00 00 00 hex
- IDF Table
- Variables over 4 Bytes Table

- DateTime Data
- Make String
- SoftwareString
- Xres Value
- Yres Value
- Strip Offsets
- Strip Counts
- Data Bytes, in order starting with Strip 0

Differences from Section 4 of Tiff 6.0

- Tag 0102 hex's Value is 16, not 4 or 8.
- Data is stored as a 16 bit unsigned Integer

*Note : All offsets are justified on 16-bit words. All pixels are justified on the words. Image strip offsets are justified on 16-bit words. An image will have the Next Strip offset 2 StripWidths * bytes past the last strip. This format is supported by a limited number of applications and is an extension to the Tiff 6.0 format.*

Index

#-A

3d check box.....	36
A to D converter	12, 34
Acquisition	49, 57, 65, 83
Adjustments button.....	29
Area Maps Tab	32
Area creation.....	32
Areas.....	83
ASCII format.....	6

B-C

BMP	6
Calibrate button	37
Calibration, CCD.....	37
CCD Based Acquisition.....	49
CCD Survey Screen.....	29
CCD Tab	12
CCD Triggers Tab.....	25
Cleaning	83
Clear button	30
Cosmic Correct check box	36
Cosmic Correction.....	83
Cursor coordinates panel	30

D-E

Dark Subtract.....	83
Dark Subtract check box	36
Detection Tab.....	14
Discrimination	78
Display format.....	44
Display tab.....	31
Excitation Profile panel	53
Expanding Spectra and Images	6

F-G

File Formats.....	6
File Operations	76
Filter panel.....	36
Filter Wheel	
auto insertion	22
set up.....	11

Filter Wheel Tab	22
Focal Plane Angle.....	22
Focusing Your CCD Detector.....	85
Gain	14
Glossary	83
GRAMS software	6
Grey/Color button.....	31
Groups	83

H-K

Hardware Config screen	9, 36, 43
Hardware Installation	7
Head orientation	13
High Voltage Table	45
History	78
Horizontal binning.....	83
Horizontal Profile button	34
HV	
Default HV.....	15
Maximum HV	14
Icons	9
Image	84
Imaging/Spectral button	29
Importing Data into GRAMS	81
Improving the signal to noise ratio of your CCD spectra	87
Info button	35
Introduction	5

L-O

Linearize check box.....	51
Live Data Screen	71
Lock-in amplifier.....	16
Lower Display Window.....	34
Monochromator Tab	21
Multifile	84
NCL Triggers Tab	24
Optical layout	10
Optical path	10
Optics Tab.....	11
Optimize button	44

P-R

Photon Counting.....	14
PMT current	14
Positive/Negative button	31
Post Processing Screen.....	75
Printer Options Tab.....	26
Quick Strips.....	32
Reading Units.....	15
Readout.....	84
Reads per point.....	44, 58
Real Time Processing panel	
CCD.....	35
scanning.....	43
ReDraw button	31
Routine panel.....	54, 61, 68

S

Scalar Operations	76
Scan Range Selection panel	41
Scanning Spectral Acquisition.....	57
Scanning Survey Screen.....	41
Scanning Time Based Acquisition.....	65
Segment.....	84

Setup tab	
CCD	30
scanning	42
Show Half Width	45
Shutters	
set up.....	11
Shutters Tab	23
safety zone	23
Signal to noise ratio (s/n).....	87
Silicon diode.....	14
Slits	
Constant band pass	21, 33
home	21
types.....	21
Software Bin check box.....	36
Software binning	84
Software Installation.....	6
Spectrum.....	84

T-Z

Tiff.....	6
Tiff 6.0 Compliance.....	91
Triggers indicator	36, 45
Vertical binning	84
Vertical Profile button.....	34
Working spectral units	11